

# The Conduction Connective: An Irreducible Extension of Intuitionistic Modal Logic IK

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## Abstract

We introduce a binary connective  $\varphi \triangleright \psi$ , called *conduction*, extending the intuitionistic modal logic IK. The satisfaction clause is:  $w \models \varphi \triangleright \psi$  iff for every  $v \geq w$ , if  $v \models \varphi$  then there exists  $u$  with  $vRu$ ,  $v \leq u$ , and  $u \models \psi$ . This composes intuitionistic implication with an existential modality over the intersection  $S = R \cap \leq$ . We show that a standalone modality  $\langle S \rangle$  fails to persist, and that  $\triangleright$  canonically resolves this. We prove irreducibility of  $\triangleright$  over IK via bisimulation. We present the Hilbert system **H-DWS** with soundness, a fully labelled sequent calculus **LS-DWS** with an annotated-challenger mechanism ensuring rule-by-rule soundness, and prove completeness for the class of  $S$ -serial bare frames and decidability via terminating proof search with a singly exponential upper bound on model size. We compare the logic with intuitionistic conditional logics, identifying conduction as a constant-selection-function specialisation of the intuitionistic might-conditional.

**Keywords:** intuitionistic modal logic; conduction connective; labelled sequent calculus; decidability; birelational Kripke semantics

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## 1 Introduction

Intuitionistic modal logic IK, introduced independently by Fischer Servi [3] and Plotkin–Stirling [7] and studied extensively by Simpson [8], is interpreted over birelational Kripke frames  $\langle W, \leq, R \rangle$  where  $\leq$  is a preorder and  $R$  is a modal accessibility relation, subject to forward confluence: if  $w \leq w'$  and  $wRu$ , then there exists  $u'$  with  $w'Ru'$  and  $u \leq u'$ .

Labelled proof theory for such logics was developed by Marin, Morales, and Straßburger [5], who presented a fully labelled sequent system for a wide family of intuitionistic modal logics.

In this paper we study the intersection

$$S = R \cap \leq,$$

the relation of *accessible refinement*, and introduce the *conduction connective*  $\varphi \triangleright \psi$ . The name **DWS** stands for **Diamond-Within-implication over S**, reflecting the construction: conduction embeds an  $S$ -indexed existential modality inside an intuitionistic implication. The language  $\mathcal{L}_{\text{DWS}}$  extends that of IK, denoted  $\mathcal{L}_{\text{IK}}$ , with this single new binary connective.

Our first contribution is semantic. We show that  $S$  need not satisfy forward confluence, even on  $S$ -serial bare frames, and hence that a standalone modality  $\langle S \rangle$  need not persist (§3).<sup>1</sup> We then identify conduction as a *coreflected residual*: the raw powerset existential along the ambient relation  $S$  is formed in the ordinary predicate fibration, and the resulting residual is coreflected back into persistent predicates.

Our second contribution is categorical. We define a concrete double category of persistent relations. Its objects are preorders, its vertical arrows are monotone maps, and its horizontal arrows are exactly those relations whose raw existential predicate transformers preserve upward-closed predicates. This is a relation-style double category, not the ordinary double category of profunctors. In it, a bare frame is precisely a non-empty preorder equipped with a horizontal endomorphism  $R$ . The relation  $S = R \cap \leq$ , however, need not be horizontal. This is the structural reason why the raw  $S$ -diamond fails and why conduction is needed. We compare this construction with ordinary profunctor collages only as a contrast: ordinary profunctorial semantics builds in stronger variance and compatibility conditions than are available for  $S$  in general, and the formal categorical structure used here is **PRel**, not the ordinary profunctor equipment.

Our third contribution is expressivity. We locate conduction between the ordinary IK modalities by the modal sandwich, prove that both inclusions are strict, identify the exact collapse boundary, prove irreducibility of  $\triangleright$  over IK, and establish truth conservativity for the IK fragment (§4).

Finally, we present the Hilbert system **H-DWS** with soundness (§5); completeness of this Hilbert system remains open (§9). We present the labelled sequent system **LS-DWS** with an annotated-challenger mechanism that makes all rules sound in isolation, and prove soundness and completeness (§6). A distinctive feature of proof search is a *redirect mechanism* that identifies labels sharing the same semantic type within a given depth level and merges them, transferring annotations and relational atoms; this keeps the label space finite and search terminating (§6.4). We prove decidability with an explicit singly exponential bound, together with the finite model property (§7). We compare the logic with intuitionistic conditional logics via a formal embedding and correspondence theorem (§8).

## 2 Preliminaries

This section fixes the Kripke, fibrational, and double-categorical notation used in the rest of the paper. The satisfaction relation is defined on arbitrary bare frames. DWS-validity is obtained by restricting validity to the class of  $S$ -serial bare frames.

**Definition 2.1** (Bare frame). A *bare frame*<sup>2</sup> is a triple  $\mathcal{F} = \langle W, \leq, R \rangle$ , where  $W \neq \emptyset$ ,  $\leq$  is a preorder on  $W$ , and  $R \subseteq W \times W$ , satisfying:

(FC) *Forward confluence*: if  $w \leq w'$  and  $wRu$ , then  $\exists u' (w'Ru' \wedge u \leq u')$ .

We write  $v \geq w$  for  $w \leq v$ . We also write  $S = R \cap \leq$ , so that  $wSu$  iff  $wRu$  and  $w \leq u$ . A bare frame is  *$S$ -serial* if:

(Ser)  $\forall w \exists u (wSu)$ .

Thus  $S$ -seriality is stronger than  $R$ -seriality ( $\forall w \exists u (wRu)$ ): every  $S$ -successor is simultaneously an  $R$ -successor and a  $\leq$ -successor. It is strictly stronger. For example, let  $W = \{a, b\}$ , let

<sup>1</sup>We use the symbol  $\triangleright$  (`\vartriangleright`) for conduction throughout. The same symbol has been used elsewhere in proof theory for the admissibility relation between rules; no connection is intended. Our choice is motivated by the visual suggestion of a directed, implication-like operator.

<sup>2</sup>We use “bare frame” to distinguish the frame  $\langle W, \leq, R \rangle$  from the full model  $\langle W, \leq, R, V \rangle$ . This is the same structure as the “birelational frame” of Simpson [8] and Fischer Servi [4], but our terminology emphasises the absence of the valuation.

$\leq$  be the identity preorder, and let  $R = \{(a, b), (b, b)\}$ . The frame is  $R$ -serial, but  $S = R \cap \leq = \{(b, b)\}$ , so  $a$  has no  $S$ -successor.

For a preorder  $(W, \leq)$ , let

$$\text{Up}_{\leq}(W) = \{A \subseteq W \mid \forall w \in W \forall v \in W ((w \in A \wedge w \leq v) \Rightarrow v \in A)\}$$

be the set of upward-closed subsets of  $W$  with respect to  $\leq$ . Equivalently,

$$\text{Up}_{\leq}(W) = \{A \subseteq W \mid \forall w \in A \forall v \in W (w \leq v \Rightarrow v \in A)\}.$$

When the preorder is fixed, we write simply  $\text{Up}(W)$ .

**Definition 2.2** (Model, language). A *model* is  $\mathcal{M} = \langle W, \leq, R, V \rangle$ , where the underlying frame is bare and

$$V: \text{Prop} \rightarrow \text{Up}_{\leq}(W)$$

assigns to each propositional variable an upward-closed subset of  $W$ . Equivalently,  $V$  is monotone: if  $w \in V(p)$  and  $w \leq w'$ , then  $w' \in V(p)$ .

The language  $\mathcal{L}_{\text{DWS}}$  extends that of IK, intuitionistic modal logic:

$$\varphi ::= p \mid \perp \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \varphi \rightarrow \psi \mid \Box \varphi \mid \Diamond \varphi \mid \varphi \triangleright \psi.$$

The binary connective  $\triangleright$  is called *conduction*. We use  $\top$  as an abbreviation for  $\perp \rightarrow \perp$ , and  $\neg\alpha$  as an abbreviation for  $\alpha \rightarrow \perp$ .

The satisfaction relation is defined as follows. The propositional clauses are the usual intuitionistic clauses:

$$\begin{aligned} w \models p & \text{ iff } w \in V(p), \\ w & \not\models \perp, \\ w \models \varphi \wedge \psi & \text{ iff } w \models \varphi \text{ and } w \models \psi, \\ w \models \varphi \vee \psi & \text{ iff } w \models \varphi \text{ or } w \models \psi, \\ w \models \varphi \rightarrow \psi & \text{ iff } \forall v \geq w (v \models \varphi \Rightarrow v \models \psi). \end{aligned}$$

The modal clauses are:

$$\begin{aligned} w \models \Box \varphi & \text{ iff } \forall v \geq w \forall u (vRu \Rightarrow u \models \varphi), \\ w \models \Diamond \varphi & \text{ iff } \exists u (wRu \wedge u \models \varphi). \end{aligned}$$

The clause for  $\Box$  is stabilised by the universal quantification over all refinements  $v \geq w$ , whereas the clause for  $\Diamond$  is the raw  $R$ -existential; its persistence uses (FC). Finally, satisfaction for conduction is given by:

$$w \models \varphi \triangleright \psi \text{ iff } \forall v \geq w (v \models \varphi \Rightarrow \exists u (vSu \wedge u \models \psi)). \quad (\text{S}\triangleright)$$

We write  $\text{Sub}(\varphi)$  for the set of subformulas of  $\varphi$ , and  $\text{md}(\varphi)$  for the *modal depth* of  $\varphi$ , defined as the maximum nesting of  $\Box$ ,  $\Diamond$ , and  $\triangleright$ . Although  $\triangleright$  is binary, it is counted as modal because its semantic clause uses  $S$ -successors. Explicitly:

$$\begin{aligned} \text{md}(p) = \text{md}(\perp) &= 0, \\ \text{md}(\alpha \circ \beta) &= \max(\text{md}(\alpha), \text{md}(\beta)) \quad (\circ \in \{\wedge, \vee, \rightarrow\}), \\ \text{md}(\Box \alpha) = \text{md}(\Diamond \alpha) &= \text{md}(\alpha) + 1, \\ \text{md}(\alpha \triangleright \beta) &= \max(\text{md}(\alpha), \text{md}(\beta)) + 1. \end{aligned}$$

**Theorem 2.3** (Persistence). *For every model  $\mathcal{M}$ , every  $w \leq w'$ , and every  $\varphi \in \mathcal{L}_{\text{DWS}}$ , if  $w \models \varphi$ , then  $w' \models \varphi$ .*

*Proof.* By induction on  $\varphi$ . For atoms, if  $w \models p$ , then  $w \in V(p)$ ; since  $V(p)$  is upward closed and  $w \leq w'$ , we obtain  $w' \in V(p)$ , hence  $w' \models p$ . The case  $\perp$  is vacuous. The cases for  $\wedge$  and  $\vee$  follow immediately from the induction hypothesis.

For implication, suppose  $w \models \alpha \rightarrow \beta$  and  $w \leq w'$ . If  $v \geq w'$  and  $v \models \alpha$ , then  $v \geq w$ , so  $v \models \beta$ . Hence  $w' \models \alpha \rightarrow \beta$ .

For  $\Box\alpha$ , suppose  $w \models \Box\alpha$  and  $w \leq w'$ . If  $v \geq w'$  and  $vRu$ , then  $v \geq w$ , hence  $u \models \alpha$ . Therefore  $w' \models \Box\alpha$ .

For  $\Diamond\alpha$ , suppose  $w \models \Diamond\alpha$ , so for some  $u$ ,  $wRu$  and  $u \models \alpha$ . Since  $w \leq w'$ , (FC) gives a world  $u'$  such that  $w'Ru'$  and  $u \leq u'$ . By the induction hypothesis,  $u' \models \alpha$ . Hence  $w' \models \Diamond\alpha$ .

For conduction, suppose  $w \models \alpha \triangleright \beta$  and  $w \leq w'$ . Let  $v \geq w'$  with  $v \models \alpha$ . By transitivity,  $v \geq w$ , so the hypothesis gives  $u$  with  $vSu$  and  $u \models \beta$ . Thus  $w' \models \alpha \triangleright \beta$ . This case does not use (FC): the universal quantification over refinements in  $(S \triangleright)$  absorbs the refinement step directly, exactly as in the clause for intuitionistic implication. Forward confluence is nevertheless essential for the language as a whole, because it is used in the  $\Diamond$ -case.  $\square$

## 2.1 Persistent predicates, fibrations, and persistent relations

We now give the categorical infrastructure used later. The point is not to introduce a second notion of validity. Rather, the Kripke clauses above are re-expressed as operations on persistent predicates, and bare frames are identified with horizontal endomorphisms in a concrete double category of persistent relations. The categorical layer is therefore truth-conservative but structurally informative: it isolates the exact stability condition for relational existentials, explains why the raw mixed relation  $S = R \cap \leq$  cannot in general be used as an ordinary intuitionistic diamond, and identifies conduction as a coreflected residual of that raw existential.

### Persistent predicates and the powerset fibration

Let  $\mathcal{F} = \langle W, \leq, R \rangle$  be a bare frame. The preorder  $(W, \leq)$  determines a thin category  $\mathcal{V}_{\mathcal{F}}$ , whose objects are worlds and with a unique arrow  $w \rightarrow v$  precisely when  $w \leq v$ . We use covariant functors  $\mathcal{V}_{\mathcal{F}} \rightarrow \mathbf{Set}$ ; equivalently,  $\mathbf{Set}^{\mathcal{V}_{\mathcal{F}}}$  may be viewed as the presheaf topos on  $\mathcal{V}_{\mathcal{F}}^{op}$ . This direction is chosen so that subterminal objects are upward closed. Indeed, a subterminal object assigns to each world either the singleton or the empty set, and functoriality along  $w \rightarrow v$  forces truth at  $w$  to persist to truth at  $v$ . Thus persistent propositions over  $\mathcal{F}$  are precisely the elements of  $\mathbf{Up}_{\leq}(W)$ .

We also use the elementary predicate-fibrational notation; see Jacobs [11] for the general categorical-logic background. We use the indexed-poset presentation of the powerset fibration; equivalently, one may take its Grothendieck construction. Let

$$\mathbf{Pred}: \mathbf{Set}^{op} \rightarrow \mathbf{Pos}$$

be the powerset indexed poset, sending a set  $X$  to  $\mathcal{P}(X)$  and a function  $f: X \rightarrow Y$  to inverse image  $f^*: \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ . The inverse-image map has a left adjoint  $\exists_f$ , given by direct image:

$$\exists_f(A) = \{y \in Y \mid \exists x \in A (f(x) = y)\}.$$

Let

$$j: \mathbf{Up}_{\leq}(W) \hookrightarrow \mathcal{P}(W)$$

be the inclusion of persistent predicates into all predicates, ordered by inclusion. This inclusion has both adjoints:

$$\mathbf{UpCl}_{\leq} \dashv j \dashv \mathbf{Int}_{\leq}.$$

Here

$$\mathbf{UpCl}_{\leq}(X) = \{v \in W \mid \exists w \in X (w \leq v)\}$$

is the upward closure of  $X$ , and

$$\text{Int}_{\leq}(X) = \{w \in W \mid \forall v \geq w, v \in X\}$$

is the largest upward-closed subset of  $X$ . Thus  $\text{Up}_{\leq}(W)$  is a reflective and coreflective subposet of  $\mathcal{P}(W)$ ; in particular,  $\text{Int}_{\leq}$  extracts the persistent core of an arbitrary predicate.

For  $A, B \in \text{Up}_{\leq}(W)$ , the Heyting operations are

$$A \wedge B = A \cap B, \quad A \vee B = A \cup B, \quad \perp = \emptyset, \quad \top = W,$$

and

$$A \Rightarrow B = \text{Int}_{\leq}((W \setminus A) \cup B) = \{w \in W \mid \forall v \geq w, v \in A \Rightarrow v \in B\}.$$

Thus intuitionistic implication is ordinary set-theoretic implication followed by coreflection into persistent predicates.

### The double category of persistent relations

We now define the double category that will carry the frame structure. The terminology is inspired by the theory of proarrows and double categories [12, 13, 14, 15], but the construction used here is elementary. We use only the strict double category defined below; in particular, we do not claim that it is an equipment or the ordinary double category of profunctors.

**Definition 2.4** (Persistent relation). Let  $(X, \leq_X)$  and  $(Y, \leq_Y)$  be preorders. A relation  $T \subseteq X \times Y$  is *persistent*, written

$$T: X \rightarrow_{\text{h}} Y,$$

if it satisfies the following source-side forward-lifting condition:

$$x \leq_X x' \wedge xTy \implies \exists y' (x'Ty' \wedge y \leq_Y y'). \quad (\text{PRel})$$

**Definition 2.5** (The double category **PRel**). The double category **PRel** is defined as follows. Objects are preorders. Vertical arrows are monotone maps. Horizontal arrows are persistent relations. Horizontal composition is ordinary relational composition:

$$(U \circ T)(x, z) \text{ iff } \exists y (xTy \wedge yUz),$$

and the horizontal identity on  $X$  is the equality relation  $=_X \subseteq X \times X$ . Given vertical arrows  $f: X \rightarrow X'$  and  $g: Y \rightarrow Y'$ , and horizontal arrows  $T: X \rightarrow_{\text{h}} Y$  and  $T': X' \rightarrow_{\text{h}} Y'$ , there is a unique cell

$$\begin{array}{ccc} X & \xrightarrow{T} & Y \\ f \downarrow & & \downarrow g \\ X' & \xrightarrow{T'} & Y' \end{array}$$

iff

$$\forall x \in X \forall y \in Y (xTy \Rightarrow f(x)T'g(y)).$$

Otherwise there is no cell with this boundary. Equivalently, the cell set for fixed boundary data is

$$\mathbf{PRel}_{f,g}(T, T') = \begin{cases} \{*\}, & \text{if } \forall x \in X \forall y \in Y (xTy \Rightarrow f(x)T'g(y)), \\ \emptyset, & \text{otherwise.} \end{cases}$$

Thus **PRel** is locally posetal at the level of cells. It is a relation-style double category: its horizontal identities are equality relations, not the hom-relations of the preorder categories. It should therefore not be confused with the ordinary double category of profunctors.

**Proposition 2.6.** **PRel** is a strict double category.

*Proof.* First, horizontal identities are persistent. If  $x \leq_X x'$  and  $x = y$ , take  $y' = x'$ . Then  $x' = y'$  and  $y = x \leq_X x' = y'$ .

Second, persistent relations are closed under horizontal composition. Let  $T: X \rightarrow_h Y$  and  $U: Y \rightarrow_h Z$  be persistent. Suppose  $x \leq_X x'$  and  $(U \circ T)(x, z)$ . Choose  $y$  with  $xTy$  and  $yUz$ . By persistence of  $T$ , choose  $y'$  with  $x'Ty'$  and  $y \leq_Y y'$ . By persistence of  $U$ , applied to  $y \leq_Y y'$  and  $yUz$ , choose  $z'$  with  $y'Uz'$  and  $z \leq_Z z'$ . Hence  $(U \circ T)(x', z')$ . Identity and associativity of horizontal composition are those of ordinary relational composition.

It remains to spell out cell composition. For fixed boundary data, there is at most one cell. Vertical composition is composition of the corresponding implications: if

$$xTy \Rightarrow f(x)T'g(y)$$

and

$$x'T'y' \Rightarrow f'(x')T''g'(y'),$$

then

$$xTy \Rightarrow f'f(x)T''g'g(y).$$

Horizontal composition is inherited from relational composition. If

$$xTy \Rightarrow f(x)T'g(y)$$

and

$$yUz \Rightarrow g(y)U'h(z),$$

then from  $(U \circ T)(x, z)$  choose  $y$  with  $xTy$  and  $yUz$ . We obtain  $f(x)T'g(y)$  and  $g(y)U'h(z)$ , hence  $(U' \circ T')(f(x), h(z))$ . Identity cells are the evident implications. Since every cell set is either empty or a singleton, the interchange law is automatic. Therefore **PREL** is a strict double category.  $\square$

**Proposition 2.7** (Bare frames as horizontal endomorphisms). *For a non-empty preorder  $(W, \leq)$  and a relation  $R \subseteq W \times W$ , the triple  $\langle W, \leq, R \rangle$  is a bare frame iff  $R: W \rightarrow_h W$  is a horizontal endomorphism in **PREL**.*

*Proof.* Unfolding Definition 2.4, persistence of  $R: W \rightarrow_h W$  says exactly:

$$w \leq w' \wedge wRu \implies \exists u' (w'Ru' \wedge u \leq u'),$$

which is (FC).  $\square$

In the definition of  $S = R \cap \leq$ , the preorder  $\leq$  is used as an ordinary ambient relation on  $W$ . It is not the horizontal identity of **PREL**, which is equality, and it need not itself be a horizontal arrow of **PREL**.

### Raw existential images and persistent transformers

Let  $T \subseteq X \times Y$  be any relation between preorders, not necessarily persistent. Let

$$T \xrightarrow{s_T} X, \quad T \xrightarrow{t_T} Y$$

be its source and target maps,  $s_T(x, y) = x$  and  $t_T(x, y) = y$ . In the powerset fibration, the raw  $T$ -existential of a predicate  $B \subseteq Y$  is

$$E_T(B) = \exists_{s_T} t_T^* B = \{x \in X \mid \exists y \in Y (xTy \wedge y \in B)\}.$$

Thus  $E_T$  is always a predicate transformer  $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ , but it need not restrict to persistent predicates.

**Theorem 2.8** (Persistent relations and predicate transformers). *Let  $T \subseteq X \times Y$  be a relation between preorders. The following are equivalent:*

- (a)  $T: X \rightarrow_h Y$  is a horizontal arrow of **PRel**.
- (b)  $T$  satisfies (PRel).
- (c) For every  $B \in \mathbf{Up}_{\leq_Y}(Y)$ , the set  $E_T(B)$  lies in  $\mathbf{Up}_{\leq_X}(X)$ .

*Proof.* The equivalence of (a) and (b) is Definition 2.4. Assume (b). Let  $B \in \mathbf{Up}_{\leq_Y}(Y)$ ,  $x \in E_T(B)$ , and  $x \leq_X x'$ . Choose  $y$  with  $xTy$  and  $y \in B$ . By (PRel), choose  $y'$  with  $x'Ty'$  and  $y \leq_Y y'$ . Since  $B$  is upward closed,  $y' \in B$ . Hence  $x' \in E_T(B)$ .

Conversely, suppose that  $E_T(B)$  is upward closed for every persistent  $B$ . Let  $x \leq_X x'$  and  $xTy$ . Put

$$\uparrow y = \{z \in Y \mid y \leq_Y z\}.$$

Then  $\uparrow y \in \mathbf{Up}_{\leq_Y}(Y)$ , and  $x \in E_T(\uparrow y)$ , witnessed by  $y$ . By upward closure,  $x' \in E_T(\uparrow y)$ . Thus there is  $y'$  with  $x'Ty'$  and  $y' \in \uparrow y$ , i.e.  $y \leq_Y y'$ . Hence  $T$  satisfies (PRel).  $\square$

**Proposition 2.9** (Functoriality of raw existentials). *For relations  $T \subseteq X \times Y$  and  $U \subseteq Y \times Z$ ,*

$$E_{U \circ T} = E_T \circ E_U$$

*as maps  $\mathcal{P}(Z) \rightarrow \mathcal{P}(X)$ . Moreover,*

$$E_{=X} = \text{id}_{\mathcal{P}(X)}.$$

*If  $T$  and  $U$  are horizontal arrows of **PRel**, then these equations restrict to persistent predicates.*

*Proof.* For  $C \subseteq Z$ ,

$$\begin{aligned} x \in E_{U \circ T}(C) & \text{ iff } \exists z ((U \circ T)(x, z) \wedge z \in C) \\ & \text{ iff } \exists y \exists z (xTy \wedge yUz \wedge z \in C) \\ & \text{ iff } \exists y (xTy \wedge y \in E_U(C)) \\ & \text{ iff } x \in E_T(E_U(C)). \end{aligned}$$

The identity equation is immediate from the equality relation. The last assertion follows from Theorem 2.8.  $\square$

**Proposition 2.10** (Cells and existential predicate transformers). *Let*

$$\begin{array}{ccc} X & \xrightarrow{T} & Y \\ f \downarrow & & \downarrow g \\ X' & \xrightarrow{T'} & Y' \end{array}$$

*be a cell in **PRel**. Then, for every predicate  $B \subseteq Y'$ ,*

$$E_T(g^*B) \subseteq f^*E_{T'}(B).$$

*If  $B \in \mathbf{Up}_{\leq_{Y'}}(Y')$ , then this inequality restricts to persistent predicates.*

*Proof.* Suppose  $x \in E_T(g^*B)$ . Then for some  $y$ ,  $xTy$  and  $g(y) \in B$ . Since the square is a cell,  $f(x)T'g(y)$ . Hence  $f(x) \in E_{T'}(B)$ , i.e.  $x \in f^*E_{T'}(B)$ . If  $B$  is persistent and  $T, T'$  are horizontal arrows, then Theorem 2.8 and monotonicity of  $f$  and  $g$  ensure that all predicates appearing in the inclusion are persistent.  $\square$

*Remark 2.11* (Morphism-level consequence). Proposition 2.10 is a genuinely structural consequence of the double-categorical formulation. A cell in **PRel** does not merely compare two relations pointwise; it induces a natural inequality between the corresponding existential predicate transformers. Thus the double-categorical layer records how accessibility relations behave under monotone change of base. This kind of morphism-level statement is not visible in the bare pointwise Kripke clauses, although it is conservative over them by Theorem 2.17 below.

For  $T = R$  on a bare frame, Theorem 2.8 explains why the raw  $R$ -diamond preserves persistent predicates:  $R$  is a horizontal endomorphism of **PRel**. For  $T = S = R \cap \leq$ , the same theorem says that a raw  $S$ -diamond would be a genuine intuitionistic diamond exactly if  $S$  were also a horizontal endomorphism of **PRel**. The next section shows that this can fail, even when the frame is  $S$ -serial.

**Corollary 2.12** (Stable and unstable raw diamonds). *Let  $\mathcal{F} = \langle W, \leq, R \rangle$  be a bare frame and let  $S = R \cap \leq$ .*

(i) *The raw  $R$ -existential restricts to a monotone endomap*

$$E_R: \mathbf{Up}_{\leq}(W) \rightarrow \mathbf{Up}_{\leq}(W).$$

*Equivalently,  $\Diamond_R$  is a genuine intuitionistic diamond on persistent predicates.*

(ii) *The raw  $S$ -existential restricts to an endomap*

$$E_S: \mathbf{Up}_{\leq}(W) \rightarrow \mathbf{Up}_{\leq}(W)$$

*iff  $S: W \rightarrow_h W$  is a horizontal endomorphism of **PRel**, equivalently iff  $S$  satisfies*

$$w \leq w' \wedge wSu \implies \exists u' (w'Su' \wedge u \leq u').$$

*Proof.* For (i), Proposition 2.7 says that  $R$  is a horizontal endomorphism of **PRel**, and Theorem 2.8 then says exactly that  $E_R$  preserves persistent predicates. For (ii), apply Theorem 2.8 to the relation  $T = S$ .  $\square$

## Semantic operations

For  $A \in \mathbf{Up}_{\leq}(W)$ , define

$$\Diamond_R A = E_R(A).$$

This is upward closed because  $R$  is persistent. Define the box by coreflecting the ordinary universal  $R$ -condition:

$$\Box_R A = \mathbf{Int}_{\leq} \{w \in W \mid \forall u (wRu \Rightarrow u \in A)\} = \{w \in W \mid \forall v \geq w \forall u (vRu \Rightarrow u \in A)\}.$$

For the mixed relation  $S = R \cap \leq$ , define the raw  $S$ -existential by

$$E_S(B) = \{v \in W \mid \exists u \in W (vSu \wedge u \in B)\}.$$

This predicate is always defined in  $\mathcal{P}(W)$ , but it may fail to be persistent because  $S$  need not be horizontal in **PRel**. Conduction is the coreflected residual of this raw powerset existential:

$$A \triangleright_S B = \mathbf{Int}_{\leq} ((W \setminus A) \cup E_S(B)).$$

Equivalently,

$$A \triangleright_S B = \{w \in W \mid \forall v \geq w, v \in A \Rightarrow v \in E_S(B)\}.$$

When  $S$  is fixed, we write simply  $A \triangleright B$ .

**Proposition 2.13** (Conduction as coreflected residual). *Let  $A, B \in \mathbf{Up}_{\leq}(W)$ . For every  $X \in \mathbf{Up}_{\leq}(W)$ ,*

$$X \subseteq A \triangleright_S B \iff X \cap A \subseteq E_S(B).$$

*Consequently,*

$$A \triangleright_S B = \bigvee \{X \in \mathbf{Up}_{\leq}(W) \mid X \cap A \subseteq E_S(B)\}.$$

*Thus  $A \triangleright_S B$  is the largest persistent predicate whose intersection with  $A$  implies the raw  $S$ -existential of  $B$ .*

*Proof.* Suppose first that  $X \subseteq A \triangleright_S B$ . Let  $w \in X \cap A$ . Since  $w \in A \triangleright_S B$ , taking  $v = w$  and using reflexivity of  $\leq$ , we obtain  $w \in E_S(B)$ . Hence  $X \cap A \subseteq E_S(B)$ .

Conversely, suppose  $X \cap A \subseteq E_S(B)$ . Let  $w \in X$ . To show  $w \in A \triangleright_S B$ , let  $v \geq w$  and suppose  $v \in A$ . Since  $X$  is upward closed and  $w \in X$ , we have  $v \in X$ . Thus  $v \in X \cap A$ , so  $v \in E_S(B)$ . Therefore  $w \in A \triangleright_S B$ . The displayed join is the union of all such persistent predicates, which is again persistent.  $\square$

When  $E_S(B)$  happens to be persistent, this coreflected residual coincides with the ordinary Heyting implication  $A \Rightarrow E_S(B)$ . In general, however,  $E_S(B)$  is not an object of  $\mathbf{Up}_{\leq}(W)$ , so  $A \triangleright_S B$  is not literally an internal Heyting implication into  $E_S(B)$ ; it is the largest persistent predicate satisfying the residual condition of Proposition 2.13.

*Remark 2.14* (Comparison with profunctor collages). A profunctor, or distributor, can be represented by its collage, which adjoins heteromorphisms between two categories; this is a standard way of packaging proarrows in double-categorical settings [16, 13, 14]. The analogy is useful: vertical arrows record refinement, while horizontal arrows record accessibility.

However, the horizontal arrows used in this paper are not ordinary profunctors on the refinement category. Ordinary profunctorial or bimodule semantics builds in functorial compatibility with the vertical structure, often leading to Kan-extension interpretations of modal operators [17]. Our bare-frame condition is different: it is the source-side existential lifting condition making  $R$  a persistent relation in **PRel**. The mixed relation  $S = R \cap \leq$  may fail even this weaker horizontal condition. Thus conduction is not introduced as the ordinary diamond of a stable profunctor. It is the coreflected residual of a raw powerset existential that may fail to be stable. The collage comparison is only contrastive; the formal categorical structure used in the proofs is **PRel**, not the ordinary profunctor equipment.

**Lemma 2.15** (Closure of the semantic operations). *If  $A, B \in \mathbf{Up}_{\leq}(W)$ , then*

$$A \Rightarrow B, \quad \Box_R A, \quad \Diamond_R A, \quad A \triangleright_S B$$

*are all elements of  $\mathbf{Up}_{\leq}(W)$ .*

*Proof.* The operations  $A \Rightarrow B$ ,  $\Box_R A$ , and  $A \triangleright_S B$  are upward closed by definition, since each is obtained by applying  $\text{Int}_{\leq}$  to an arbitrary predicate. For  $\Diamond_R A = E_R(A)$ , use Theorem 2.8 with  $T = R$ , together with Proposition 2.7.  $\square$

**Definition 2.16** (Categorical interpretation). Let  $\mathcal{M} = \langle W, \leq, R, V \rangle$  be a model. The categorical interpretation

$$\llbracket - \rrbracket_{\mathcal{M}}: \mathcal{L}_{\text{DWS}} \rightarrow \mathbf{Up}_{\leq}(W)$$

is defined recursively by:

$$\begin{aligned}
\llbracket p \rrbracket_{\mathcal{M}} &= V(p), \\
\llbracket \perp \rrbracket_{\mathcal{M}} &= \emptyset, \\
\llbracket \varphi \wedge \psi \rrbracket_{\mathcal{M}} &= \llbracket \varphi \rrbracket_{\mathcal{M}} \cap \llbracket \psi \rrbracket_{\mathcal{M}}, \\
\llbracket \varphi \vee \psi \rrbracket_{\mathcal{M}} &= \llbracket \varphi \rrbracket_{\mathcal{M}} \cup \llbracket \psi \rrbracket_{\mathcal{M}}, \\
\llbracket \varphi \rightarrow \psi \rrbracket_{\mathcal{M}} &= \llbracket \varphi \rrbracket_{\mathcal{M}} \Rightarrow \llbracket \psi \rrbracket_{\mathcal{M}}, \\
\llbracket \Box \varphi \rrbracket_{\mathcal{M}} &= \Box_R \llbracket \varphi \rrbracket_{\mathcal{M}}, \\
\llbracket \Diamond \varphi \rrbracket_{\mathcal{M}} &= \Diamond_R \llbracket \varphi \rrbracket_{\mathcal{M}}, \\
\llbracket \varphi \triangleright \psi \rrbracket_{\mathcal{M}} &= \llbracket \varphi \rrbracket_{\mathcal{M}} \triangleright_S \llbracket \psi \rrbracket_{\mathcal{M}}.
\end{aligned}$$

The interpretation is well-defined by structural induction using Lemma 2.15.

**Theorem 2.17** (Equivalence with Kripke semantics). *For every model  $\mathcal{M} = \langle W, \leq, R, V \rangle$ , every  $\varphi \in \mathcal{L}_{\text{DWS}}$ , and every  $w \in W$ ,*

$$\mathcal{M}, w \models \varphi \iff w \in \llbracket \varphi \rrbracket_{\mathcal{M}}.$$

*Proof.* By induction on  $\varphi$ . The atomic, bottom, conjunction, and disjunction cases are immediate from the definitions. The implication case is exactly the definition of  $A \Rightarrow B$ . For  $\Box$ ,  $w \in \Box_R \llbracket \alpha \rrbracket_{\mathcal{M}}$  iff

$$\forall v \geq w \forall u (vRu \Rightarrow u \in \llbracket \alpha \rrbracket_{\mathcal{M}}),$$

which is the Kripke clause by the induction hypothesis. The case for  $\Diamond$  is exactly the definition of  $E_R$ , again using the induction hypothesis.

For conduction, using the induction hypothesis,

$$w \in \llbracket \alpha \rrbracket_{\mathcal{M}} \triangleright_S \llbracket \beta \rrbracket_{\mathcal{M}}$$

holds iff

$$\forall v \geq w, v \in \llbracket \alpha \rrbracket_{\mathcal{M}} \Rightarrow v \in E_S(\llbracket \beta \rrbracket_{\mathcal{M}}).$$

Expanding  $E_S$ , this is equivalent to

$$\forall v \geq w, \mathcal{M}, v \models \alpha \Rightarrow \exists u (vSu \wedge \mathcal{M}, u \models \beta),$$

which is precisely the Kripke clause for  $\alpha \triangleright \beta$ .  $\square$

**Corollary 2.18** (Agreement of validities). *Let  $\mathcal{K}$  be any class of bare frames, for example the class of  $S$ -serial bare frames. By Kripke validity on  $\mathcal{K}$  we mean validity in every model based on a frame in  $\mathcal{K}$ , at every world. Equivalently,*

$$\forall \mathcal{F} = \langle W, \leq, R \rangle \in \mathcal{K} \forall V: \text{Prop} \rightarrow \text{Up}_{\leq}(W) \forall w \in W, \quad \langle W, \leq, R, V \rangle, w \models \varphi.$$

*A formula  $\varphi \in \mathcal{L}_{\text{DWS}}$  is valid on  $\mathcal{K}$  in this Kripke sense iff, for every model  $\mathcal{M}$  based on a frame in  $\mathcal{K}$ ,*

$$\llbracket \varphi \rrbracket_{\mathcal{M}} = W.$$

*In particular, DWS-validity is obtained by taking  $\mathcal{K}$  to be the class of  $S$ -serial bare frames.*

*Proof.* Immediate from Theorem 2.17.  $\square$

**Remark 2.19** (Truth-conservative but structurally informative). Corollary 2.18 shows that the categorical interpretation does not change the truth conditions or the resulting notion of validity. Its contribution is structural rather than truth-theoretic. It turns frame conditions into statements about predicate transformers, separates stable existentials from unstable raw existentials, and gives conduction the universal property of a coreflected residual. These reformulations are used below to locate conduction between  $\Box(\varphi \rightarrow \psi)$  and  $\varphi \rightarrow \Diamond\psi$ , and to identify the exact collapse boundary at which conduction becomes  $\varphi \rightarrow \Diamond\psi$ .

**Proposition 2.20** (Predicate-transformer form of  $S$ -seriality). *For a bare frame  $\mathcal{F} = \langle W, \leq, R \rangle$ , the following are equivalent:*

- (a)  $\mathcal{F}$  is  $S$ -serial.
- (b) For every  $A \in \mathbf{Up}_{\leq}(W)$ ,

$$A \subseteq E_R(A).$$

*Equivalently,  $S$ -seriality is the pointwise predicate-transformer inequality*

$$\text{id}_{\mathbf{Up}_{\leq}(W)} \leq \Diamond_R.$$

*Consequently, the axiom scheme  $\alpha \rightarrow \Diamond \alpha$  is the formula-level expression of this inequality.*

*Proof.* Assume first that  $\mathcal{F}$  is  $S$ -serial, and let  $A \in \mathbf{Up}_{\leq}(W)$ . If  $w \in A$ , choose  $u$  with  $wSu$ . Then  $wRu$  and  $w \leq u$ . Since  $A$  is upward closed,  $u \in A$ . Hence  $w \in E_R(A)$ , so  $A \subseteq E_R(A)$ .

Conversely, assume that  $A \subseteq E_R(A)$  for every persistent predicate  $A$ . Fix  $w \in W$ , and put

$$\uparrow w = \{x \in W \mid w \leq x\}.$$

Then  $\uparrow w \in \mathbf{Up}_{\leq}(W)$  and  $w \in \uparrow w$ . By the assumed inclusion,  $w \in E_R(\uparrow w)$ . Thus there is  $u \in W$  such that  $wRu$  and  $u \in \uparrow w$ , i.e.  $w \leq u$ . Hence  $wSu$ . Since  $w$  was arbitrary,  $\mathcal{F}$  is  $S$ -serial.

The final statement follows by applying Theorem 2.17 to the interpretation of  $\alpha \rightarrow \Diamond \alpha$ .  $\square$

*Remark 2.21* (Internal type-theoretic reading). The fibrational semantics also suggests a type-theoretic reading of conduction. If propositions are regarded as world-indexed types, then the clause for  $A \triangleright B$  at a world  $w$  has the dependent  $\Pi\Sigma$ -shape

$$\prod_{v:W} (w \leq v) \rightarrow A(v) \rightarrow \sum_{u:W} (vRu \times v \leq u \times B(u)).$$

The outer dependent product expresses the intuitionistic refinement component, while the inner dependent sum expresses the existential  $S$ -transition. This display is only an informal internal-language reading of the semantics. A formal proof-term calculus would require additional choices, for example about proof-irrelevance for the relations  $\leq$  and  $R$ , and about transport along refinement arrows. The present paper does not develop such a calculus.

### 3 Why Conduction

We now explain why the connective  $\triangleright$  is needed. Since  $S = R \cap \leq$ , it is natural to ask whether one could simply add a standalone diamond modality along the mixed relation  $S$ . The answer is negative. By Theorem 2.8, a raw  $S$ -diamond would be stable exactly when  $S$  were a horizontal endomorphism in  $\mathbf{PRel}$ . The following example shows that this condition is not guaranteed by the bare-frame axiom (FC), even in  $S$ -serial frames.

**Proposition 3.1** (The mixed relation need not be horizontal). *There exists an  $S$ -serial bare frame  $\langle W, \leq, R \rangle$  such that  $R: W \rightarrow_h W$  is horizontal in  $\mathbf{PRel}$ , but  $S = R \cap \leq$  is not horizontal in  $\mathbf{PRel}$ . Equivalently,  $S$  does not satisfy  $S$ -FC, namely the condition*

$$w \leq w' \wedge wSu \implies \exists u' (w'Su' \wedge u \leq u').$$

*Proof.* Let  $W = \{a, b, c, d\}$ . Define  $\leq$  as the reflexive-transitive closure of

$$\{(a, b), (a, c), (c, d)\}.$$

Thus

$$a \leq b, \quad a \leq c, \quad c \leq d, \quad a \leq d,$$

together with the reflexive pairs; in particular

$$b \not\leq c, \quad b \not\leq d, \quad c \not\leq b.$$

Set

$$R = \{(a, c), (b, b), (b, d), (c, d), (d, d)\}.$$

We verify (FC) for every relevant pair  $(w \leq w', wRu)$ :

| $w \leq w'$ | $wRu$ | witness $u'$ | check           |
|-------------|-------|--------------|-----------------|
| $a \leq a$  | $aRc$ | $c$          | $aRc, c \leq c$ |
| $a \leq b$  | $aRc$ | $d$          | $bRd, c \leq d$ |
| $a \leq c$  | $aRc$ | $d$          | $cRd, c \leq d$ |
| $a \leq d$  | $aRc$ | $d$          | $dRd, c \leq d$ |
| $b \leq b$  | $bRb$ | $b$          | $bRb, b \leq b$ |
| $b \leq b$  | $bRd$ | $d$          | $bRd, d \leq d$ |
| $c \leq c$  | $cRd$ | $d$          | $cRd, d \leq d$ |
| $c \leq d$  | $cRd$ | $d$          | $dRd, d \leq d$ |
| $d \leq d$  | $dRd$ | $d$          | $dRd, d \leq d$ |

Hence  $\langle W, \leq, R \rangle$  is a bare frame, equivalently  $R$  is horizontal in **PRel**.

The intersection is

$$S = \{(a, c), (b, b), (c, d), (d, d)\},$$

noting that  $bRd$  but  $b \not\leq d$ , so  $(b, d) \notin S$ . Moreover, the frame is  $S$ -serial:

$$aSc, \quad bSb, \quad cSd, \quad dSd.$$

However,  $S$ -FC fails. From  $a \leq b$  and  $aSc$ , we would need some  $u'$  such that  $bSu'$  and  $c \leq u'$ . The only  $S$ -successor of  $b$  is  $b$  itself, and  $c \not\leq b$ . Hence no such witness exists. By Theorem 2.8,  $S$  is not a horizontal endomorphism of **PRel**.  $\square$

**Corollary 3.2** (The raw  $S$ -existential is not stable). *The operation  $E_S$  is not, in general, an endomap on  $\mathbf{Up}_{\leq}(W)$ , even over  $S$ -serial bare frames. Consequently, a standalone modality  $\langle S \rangle \psi$ , with clause*

$$w \models \langle S \rangle \psi \quad \text{iff} \quad \exists u (wSu \wedge u \models \psi),$$

*does not in general persist in  $S$ -serial bare frames.*

*Proof.* Use the frame of Proposition 3.1. Let

$$B = \{c, d\}.$$

Then  $B \in \mathbf{Up}_{\leq}(W)$ . We have  $a \in E_S(B)$ , witnessed by  $aSc$  and  $c \in B$ . But  $b \notin E_S(B)$ , because the only  $S$ -successor of  $b$  is  $b$ , and  $b \notin B$ . Indeed,  $E_S(B) = \{a, c, d\}$ . Since  $a \leq b$ , the set  $E_S(B)$  is not upward closed.

Equivalently, take a valuation  $V(p) = \{c, d\}$ . Then  $a \models \langle S \rangle p$ , witnessed by  $c$ , but  $b \not\models \langle S \rangle p$ . Yet  $a \leq b$ . Hence  $\langle S \rangle p$  is not persistent.  $\square$

*Remark 3.3* (Double-categorical failure). The example has a precise double-categorical interpretation. The relation  $R$  is a horizontal endomorphism in **PRel**: from  $a \leq b$  and  $aRc$ , forward confluence supplies the square

$$\begin{array}{ccc} a & \xrightarrow{R} & c \\ \downarrow \leq & & \downarrow \leq \\ b & \xrightarrow{R} & d. \end{array}$$

However, the lower horizontal edge  $bRd$  is not an  $S$ -edge, because  $b \not\leq d$ . Thus the square completion takes place in the ambient horizontal relation  $R$ , but it leaves the mixed subrelation  $S = R \cap \leq$ . In the formal language of **PRel**,  $R$  is a horizontal endomorphism whereas  $S$  is only an ambient subrelation of  $R$ , not necessarily a horizontal arrow.

*Remark 3.4* (Why implication is needed). The connective  $\triangleright$  repairs the failure at the predicate level, not at the horizontal-arrow level. It does not make  $S$  a horizontal arrow of **PRel**. Instead it first forms the raw predicate  $E_S(B)$  in the ordinary powerset fibration and then coreflects the corresponding residual into persistent predicates:

$$A \triangleright_S B = \text{Int}_{\leq}((W \setminus A) \cup E_S(B)).$$

Equivalently, by Proposition 2.13, it is the largest persistent  $X$  such that

$$X \cap A \subseteq E_S(B).$$

Thus conduction is the coreflected residual of a generally unstable raw powerset existential. In particular,  $A \triangleright_S B$  is persistent by construction even when  $E_S(B)$  is not.

*Remark 3.5* (Unfolding  $S$ -FC). Since  $wSu$  means  $wRu$  and  $w \leq u$ , the condition that  $S = R \cap \leq$  be horizontal is exactly the following strengthened form of forward confluence:

$$w \leq w' \wedge wRu \wedge w \leq u \implies \exists u' (w'Ru' \wedge u \leq u' \wedge w' \leq u'). \quad (1)$$

Indeed, the extra conjunct  $w' \leq u'$  is precisely what is needed to turn the  $R$ -edge  $w'Ru'$  supplied by (FC) into an  $S$ -edge  $w'Su'$ . In the frame above, (FC) completes the instance  $a \leq b$  and  $aRc$  by the witness  $d$ , since  $bRd$  and  $c \leq d$ . But  $b \not\leq d$ , so this completion leaves  $S$ .

**Proposition 3.6** (Sufficient and insufficient conditions for  $S$ -forward confluence). *For bare frames  $\langle W, \leq, R \rangle$ , the following hold.*

- (i) *If  $R \subseteq \leq$ , then  $S = R$ , and  $S$  is a horizontal endomorphism in **PRel**.*
- (ii) *Adding the target-side backward confluence condition*

$$wRu \wedge u \leq u' \implies \exists w' (w \leq w' \wedge w'Ru') \quad (\text{BC})$$

*to (FC) does not force  $S$  to be a horizontal endomorphism in **PRel**, even in  $S$ -serial frames.*

*Proof.* For (i), if  $R \subseteq \leq$ , then  $S = R \cap \leq = R$ . Suppose  $w \leq w'$  and  $wSu$ . Since  $S = R$ , we have  $wRu$ . By (FC), there is  $u'$  with  $w'Ru'$  and  $u \leq u'$ . Since  $R \subseteq \leq$ , the edge  $w'Ru'$  also gives  $w' \leq u'$ , so  $w'Su'$ . Hence  $S$ -FC holds; equivalently,  $S$  is a horizontal endomorphism in **PRel**.

For (ii), use the frame of Proposition 3.1. It is already an  $S$ -serial bare frame, and Proposition 3.1 shows that  $S$  is not a horizontal endomorphism. It remains only to check (BC). The relevant pairs  $(wRu, u \leq u')$  are:

| $wRu$ | $u \leq u'$ | witness $w'$ | check           |
|-------|-------------|--------------|-----------------|
| $aRc$ | $c \leq c$  | $a$          | $a \leq a, aRc$ |
| $aRc$ | $c \leq d$  | $b$          | $a \leq b, bRd$ |
| $bRb$ | $b \leq b$  | $b$          | $b \leq b, bRb$ |
| $bRd$ | $d \leq d$  | $b$          | $b \leq b, bRd$ |
| $cRd$ | $d \leq d$  | $c$          | $c \leq c, cRd$ |
| $dRd$ | $d \leq d$  | $d$          | $d \leq d, dRd$ |

The only case involving a non-reflexive codomain extension is  $aRc$  together with  $c \leq d$ ; it is witnessed by  $w' = b$ , since  $a \leq b$  and  $bRd$ . In all remaining cases the codomain inequality is reflexive, and the original  $R$ -edge itself supplies the witness. Thus (BC) holds, while Proposition 3.1 shows that  $S$  is not a horizontal endomorphism in **PRel**.  $\square$

*Remark 3.7* (Structural sufficient conditions). The condition  $R \subseteq \leq$  is a simple sufficient condition for  $S$ -FC, but it is very strong: it collapses every modal transition into a refinement transition. Equation (1) gives the exact condition needed for  $S = R \cap \leq$  to be horizontal. Thus the remaining task is not to formulate  $S$ -FC by brute-force expansion, but to identify natural structural assumptions on the interaction between  $R$  and  $\leq$  that guarantee (1) without requiring  $R \subseteq \leq$ .

## 4 The Modal Sandwich, Irreducibility, and Truth Conservativity

This section locates conduction between the ordinary IK modal operations, shows that both bounds are strict, identifies the exact collapse boundary, and proves that conduction is not definable in the IK fragment. We use the categorical notation of §2.1. Thus, for  $A = \llbracket \varphi \rrbracket_{\mathcal{M}}$  and  $B = \llbracket \psi \rrbracket_{\mathcal{M}}$ ,

$$\begin{aligned}\llbracket \Box(\varphi \rightarrow \psi) \rrbracket_{\mathcal{M}} &= \Box_R(A \Rightarrow B), \\ \llbracket \varphi \triangleright \psi \rrbracket_{\mathcal{M}} &= A \triangleright_S B,\end{aligned}$$

and

$$\llbracket \varphi \rightarrow \Diamond \psi \rrbracket_{\mathcal{M}} = A \Rightarrow E_R(B).$$

**Proposition 4.1** (Modal sandwich). *Over  $S$ -serial bare frames, for all  $A, B \in \mathbf{Up}_{\leq}(W)$ ,*

$$\Box_R(A \Rightarrow B) \subseteq A \triangleright_S B \subseteq A \Rightarrow E_R(B).$$

*Equivalently, the following formulas are valid over  $S$ -serial bare frames:*

$$\begin{aligned}\models_{S\text{-ser}} \Box(\varphi \rightarrow \psi) \rightarrow (\varphi \triangleright \psi), \\ \models_{S\text{-ser}} (\varphi \triangleright \psi) \rightarrow (\varphi \rightarrow \Diamond \psi).\end{aligned}$$

*Proof.* For the left inclusion, suppose  $w \in \Box_R(A \Rightarrow B)$ . We show that  $w \in A \triangleright_S B$ . Let  $v \geq w$  and suppose  $v \in A$ . By (Ser), choose  $u$  with  $vSu$ . Then  $vRu$  and  $v \leq u$ . Since  $A$  is upward closed,  $u \in A$ . Since  $w \in \Box_R(A \Rightarrow B)$ , the facts  $v \geq w$  and  $vRu$  give  $u \in A \Rightarrow B$ . Hence  $u \in B$ . Thus  $v \in E_S(B)$ , and therefore  $w \in A \triangleright_S B$ .

For the right inclusion, suppose  $w \in A \triangleright_S B$ . We show that  $w \in A \Rightarrow E_R(B)$ . Let  $v \geq w$  and suppose  $v \in A$ . Since  $w \in A \triangleright_S B$ , there is  $u$  with  $vSu$  and  $u \in B$ . Since  $S \subseteq R$ , we have  $vRu$ , and hence  $v \in E_R(B)$ . Thus  $w \in A \Rightarrow E_R(B)$ . The right inclusion uses only  $S \subseteq R$ , not  $S$ -seriality.

The formula-level validities follow by applying the categorical interpretation and Theorem 2.17.  $\square$

**Proposition 4.2** (Strictness). *Neither inclusion in Proposition 4.1 reverses over the class of  $S$ -serial bare frames.*

*Proof.* The right inclusion does not reverse. Let  $W = \{w, t\}$ , let  $\leq$  be the identity preorder, and let

$$R = \{(w, w), (w, t), (t, t)\}.$$

Let  $V(p) = \{w, t\}$  and  $V(q) = \{t\}$ . These valuations are upward closed because  $\leq$  is the identity preorder. Then

$$S = R \cap \leq = \{(w, w), (t, t)\}.$$

The frame is bare, since  $\leq$  is the identity preorder and (FC) is immediate. It is  $S$ -serial, witnessed by the reflexive  $S$ -loops.

At  $w$ , we have  $w \models p \rightarrow \Diamond q$ : the only  $\leq$ -extension of  $w$  is  $w$  itself, and  $wRt$  with  $t \models q$ . But  $w \not\models p \triangleright q$ , since at  $v = w$  the only  $S$ -successor is  $w$ , and  $w \not\models q$ . Thus  $p \rightarrow \Diamond q$  does not imply  $p \triangleright q$ .

The left inclusion does not reverse. Let  $W = \{w, s, t\}$ . Let  $\leq$  be the reflexive-transitive closure of

$$\{(w, s), (t, s)\},$$

and let

$$R = \{(w, s), (w, t), (s, s), (t, s)\}.$$

Let  $V(p) = W$  and  $V(q) = \{s\}$ . These valuations are upward closed; in particular,  $s$  has no proper  $\leq$ -successors. Then

$$S = R \cap \leq = \{(w, s), (s, s), (t, s)\}.$$

The frame is  $S$ -serial, witnessed by

$$wSs, \quad sSs, \quad tSs.$$

It is bare. The only nontrivial (FC) checks are:

$$w \leq s \wedge wRs \rightsquigarrow sRs \wedge s \leq s,$$

$$w \leq s \wedge wRt \rightsquigarrow sRs \wedge t \leq s,$$

and

$$t \leq s \wedge tRs \rightsquigarrow sRs \wedge s \leq s.$$

All omitted (FC) cases are reflexive.

We have  $w \models p \triangleright q$ . Indeed, the  $\leq$ -extensions of  $w$  are  $w$  and  $s$ . At  $v = w$ , the  $S$ -successor  $s$  satisfies  $q$ ; at  $v = s$ , the  $S$ -successor  $s$  again satisfies  $q$ . But  $w \not\models \Box(p \rightarrow q)$ : since  $wRt$ , it would be necessary that  $t \models p \rightarrow q$ , whereas  $t \models p$  and  $t \not\models q$ . Thus  $p \triangleright q$  does not imply  $\Box(p \rightarrow q)$ .  $\square$

*Remark 4.3* (Why the sandwich is strict). The two counterexamples exhibit two different double-categorical phenomena. The formula  $p \rightarrow \Diamond q$  only asks for an ambient  $R$ -edge into a  $q$ -world; it does not ask whether that edge is also a refinement edge. By contrast,  $p \triangleright q$  asks for an  $S$ -edge, i.e. an ambient  $R$ -edge whose target is also a refinement of its source. This does not presuppose that  $S$  is itself a horizontal arrow of **PRel**. Conversely,  $\Box(p \rightarrow q)$  is universal over all ambient  $R$ -successors, while conduction only asks for the existence of an appropriate  $S$ -successor. Extra  $R$ -edges outside  $S$  can falsify the box while leaving conduction true.

**Definition 4.4** ( $S$ -cofinality in  $R$ ). A bare frame satisfies  $S$ -cofinality in  $R$  if every  $R$ -edge can be covered, at the same source, by an  $S$ -edge whose target is an upward refinement of the original target:

$$\forall v \forall u (vRu \Rightarrow \exists u' (vSu' \wedge u \leq u')). \quad (\text{Cof})$$

Equivalently,  $S$  is target-cofinal inside each source fibre of  $R$ .

**Proposition 4.5** (Collapse boundary). For a bare frame  $\mathcal{F} = \langle W, \leq, R \rangle$ , the following are equivalent:

- (a) For all models on  $\mathcal{F}$ , all worlds, and all formulas  $\varphi, \psi$ ,

$$\varphi \triangleright \psi \leftrightarrow \varphi \rightarrow \Diamond \psi$$

holds.

(b) For every  $B \in \mathbf{Up}_{\leq}(W)$ ,

$$E_R(B) \subseteq E_S(B).$$

Since  $S \subseteq R$ , this is equivalently  $E_R(B) = E_S(B)$  for all persistent  $B$ .

(c)  $\mathcal{F}$  satisfies  $S$ -cofinality in  $R$ , i.e. (Cof) holds.

Moreover, the collapse is already tested by the scheme

$$\top \triangleright p \leftrightarrow \top \rightarrow \Diamond p :$$

it holds everywhere for all valuations of  $p$  iff (Cof) holds. No  $S$ -seriality assumption is needed for this equivalence; when one restricts to DWS-validity,  $S$ -seriality is imposed independently.

*Proof.* (c) $\Rightarrow$ (b). Let  $B \in \mathbf{Up}_{\leq}(W)$  and suppose  $v \in E_R(B)$ . Then for some  $u$ ,  $vRu$  and  $u \in B$ . By (Cof), choose  $u'$  such that  $vSu'$  and  $u \leq u'$ . Since  $B$  is upward closed,  $u' \in B$ . Hence  $v \in E_S(B)$ .

(b) $\Rightarrow$ (c). Suppose  $vRu$ . Put

$$\uparrow u = \{x \in W \mid u \leq x\}.$$

Then  $\uparrow u \in \mathbf{Up}_{\leq}(W)$ , and  $v \in E_R(\uparrow u)$ , witnessed by  $u$ . By (b),  $v \in E_S(\uparrow u)$ . Hence there exists  $u'$  such that  $vSu'$  and  $u' \in \uparrow u$ , i.e.  $u \leq u'$ . Thus (Cof) holds.

(b) $\Rightarrow$ (a). Let  $A, B \in \mathbf{Up}_{\leq}(W)$ . By (b) and  $S \subseteq R$ , we have  $E_S(B) = E_R(B)$ . The set  $E_R(B) = \Diamond_R B$  is upward closed by the bare-frame condition (FC), equivalently by Theorem 2.8. Hence

$$A \triangleright_S B = \mathbf{Int}_{\leq}((W \setminus A) \cup E_S(B)) = \mathbf{Int}_{\leq}((W \setminus A) \cup E_R(B)) = A \Rightarrow E_R(B).$$

By the categorical interpretation, this is exactly the equivalence between  $\varphi \triangleright \psi$  and  $\varphi \rightarrow \Diamond \psi$ .

(a) $\Rightarrow$ (b). It is enough to use the test scheme. Let  $B \in \mathbf{Up}_{\leq}(W)$ , and interpret  $p$  by  $V(p) = B$ . Under (a),

$$\top \triangleright p \leftrightarrow \top \rightarrow \Diamond p$$

holds everywhere. The interpretation of  $\top \rightarrow \Diamond p$  is  $E_R(B)$ , since  $E_R(B)$  is upward closed by (FC). The interpretation of  $\top \triangleright p$  is  $\mathbf{Int}_{\leq}(E_S(B))$ . Hence

$$E_R(B) = \mathbf{Int}_{\leq}(E_S(B)) \subseteq E_S(B),$$

so (b) holds. □

*Remark 4.6* (Categorical form of the collapse). Proposition 4.5 says that the collapse of conduction to  $\varphi \rightarrow \Diamond \psi$  is not merely a syntactic accident. It is precisely the equality of the two existential predicate transformers  $E_S$  and  $E_R$  on persistent predicates. Since  $S \subseteq R$ , one always has  $E_S(B) \subseteq E_R(B)$  in the ambient powerset  $\mathcal{P}(W)$ . The substantive content of (Cof) is the reverse inclusion on upward-closed  $B$ : every  $R$ -witness into a persistent predicate can be replaced by an  $S$ -witness into the same predicate. Thus the collapse boundary is best understood as equality of predicate transformers, even though it is equivalently captured by the pointwise cofinality condition (Cof).

**Example 4.7** (Cofinality without inclusion). Condition (Cof) can hold even when  $R \not\subseteq \leq$ . Let

$$W = \{a, b, c\},$$

let

$$\leq = \{(a, a), (b, b), (c, c), (a, c), (b, c)\},$$

and let

$$R = \{(a, b), (a, c), (b, c), (c, c)\}.$$

Then

$$S = R \cap \leq = \{(a, c), (b, c), (c, c)\}.$$

The frame is  $S$ -serial, witnessed by

$$aSc, \quad bSc, \quad cSc.$$

It is bare. The only nontrivial (FC) checks are:

$$a \leq c \wedge aRb \rightsquigarrow cRc \wedge b \leq c,$$

$$a \leq c \wedge aRc \rightsquigarrow cRc \wedge c \leq c,$$

and

$$b \leq c \wedge bRc \rightsquigarrow cRc \wedge c \leq c.$$

All omitted (FC) cases are reflexive. Moreover,  $aRb$  but  $a \not\leq b$ , so  $R \not\subseteq \leq$ . Nevertheless (Cof) holds:

$$(a, b) : aSc \wedge b \leq c,$$

$$(a, c) : aSc \wedge c \leq c,$$

$$(b, c) : bSc \wedge c \leq c,$$

$$(c, c) : cSc \wedge c \leq c.$$

Thus collapse to  $\varphi \rightarrow \Diamond\psi$  does not require  $R \subseteq \leq$ ; it requires target-cofinality of  $S$  inside  $R$ .

**Definition 4.8** (IK-reduct). Let

$$\mathcal{M} = \langle W, \leq, R, V \rangle$$

be a DWS model. The *IK-reduct* of  $\mathcal{M}$ , written

$$\mathcal{M}^{\text{IK}},$$

is the same structure  $\langle W, \leq, R, V \rangle$ , regarded as a model only for the IK-language  $\mathcal{L}_{\text{IK}}$ . Its satisfaction relation is given by the clauses for

$$p, \quad \perp, \quad \wedge, \quad \vee, \quad \rightarrow, \quad \Box, \quad \Diamond$$

from Definition 2.2; no clause for  $\triangleright$  is present in the reduct.

**Definition 4.9** (IK-bisimulation for reducts). Let  $\mathcal{M}$  and  $\mathcal{N}$  be DWS models. An *IK-bisimulation* between their IK-reducts

$$\mathcal{M}^{\text{IK}} \quad \text{and} \quad \mathcal{N}^{\text{IK}}$$

is a relation  $Z \subseteq W_M \times W_N$  such that, for every  $(x, y) \in Z$ :

(i) *Atomic harmony*: for every propositional variable  $r$ ,

$$x \in V_M(r) \quad \text{iff} \quad y \in V_N(r).$$

(ii) *Zig- $\leq$* : if  $x \leq_M x'$ , then there is  $y'$  with  $y \leq_N y'$  and  $(x', y') \in Z$ .

(iii) *Zag- $\leq$* : if  $y \leq_N y'$ , then there is  $x'$  with  $x \leq_M x'$  and  $(x', y') \in Z$ .

(iv) *Zig- $R$* : if  $xR_M x'$ , then there is  $y'$  with  $yR_N y'$  and  $(x', y') \in Z$ .

(v) *Zag-R*: if  $yR_N y'$ , then there is  $x'$  with  $xR_M x'$  and  $(x', y') \in Z$ .

**Theorem 4.10** (Bisimulation invariance for the IK fragment). *If  $Z$  is an IK-bisimulation between the IK-reducts of  $\mathcal{M}$  and  $\mathcal{N}$ , and  $(w, a) \in Z$ , then for every  $\varphi \in \mathcal{L}_{\text{IK}}$ ,*

$$\mathcal{M}, w \models \varphi \iff \mathcal{N}, a \models \varphi.$$

*Proof.* The proof is by induction on IK-formulas. The atomic case is atomic harmony, the  $\perp$ -case is vacuous, and the Boolean cases are immediate from the induction hypothesis. We spell out the modal and implicational cases, since the clause for  $\Box$  combines refinement and accessibility.

For implication, suppose first that  $(x, y) \in Z$  and  $\mathcal{M}, x \models \alpha \rightarrow \beta$ . To show  $\mathcal{N}, y \models \alpha \rightarrow \beta$ , let  $y \leq_N y_1$  and assume  $\mathcal{N}, y_1 \models \alpha$ . By *Zag- $\leq$* , choose  $x_1$  such that  $x \leq_M x_1$  and  $(x_1, y_1) \in Z$ . By the induction hypothesis,  $\mathcal{M}, x_1 \models \alpha$ . Since  $\mathcal{M}, x \models \alpha \rightarrow \beta$ , we obtain  $\mathcal{M}, x_1 \models \beta$ , and another use of the induction hypothesis gives  $\mathcal{N}, y_1 \models \beta$ . The converse direction is symmetric and uses *Zig- $\leq$* .

For  $\Diamond\alpha$ , suppose  $(x, y) \in Z$  and  $\mathcal{M}, x \models \Diamond\alpha$ . Choose  $x_1$  with  $xR_M x_1$  and  $\mathcal{M}, x_1 \models \alpha$ . By *Zig-R*, choose  $y_1$  with  $yR_N y_1$  and  $(x_1, y_1) \in Z$ . The induction hypothesis gives  $\mathcal{N}, y_1 \models \alpha$ , hence  $\mathcal{N}, y \models \Diamond\alpha$ . The converse direction uses *Zag-R*.

For  $\Box\alpha$ , suppose  $(x, y) \in Z$  and  $\mathcal{M}, x \models \Box\alpha$ . To show  $\mathcal{N}, y \models \Box\alpha$ , let  $y \leq_N y_1$  and  $y_1 R_N y_2$ . By *Zag- $\leq$* , choose  $x_1$  with  $x \leq_M x_1$  and  $(x_1, y_1) \in Z$ . By *Zag-R*, choose  $x_2$  with  $x_1 R_M x_2$  and  $(x_2, y_2) \in Z$ . Since  $\mathcal{M}, x \models \Box\alpha$ , we have  $\mathcal{M}, x_2 \models \alpha$ . By the induction hypothesis,  $\mathcal{N}, y_2 \models \alpha$ . Thus  $\mathcal{N}, y \models \Box\alpha$ . The converse direction is the same argument with *Zig- $\leq$*  followed by *Zig-R*.  $\square$

**Theorem 4.11** (Irreducibility over the IK fragment). *There is no formula of  $\mathcal{L}_{\text{IK}}$  equivalent to  $p \triangleright q$  over the class of  $S$ -serial bare frames.*

*Proof.* We exhibit two  $S$ -serial bare models  $\mathcal{M}$  and  $\mathcal{N}$ , and an IK-bisimulation  $Z$  between their IK-reducts, such that

$$\mathcal{M}, w \models p \triangleright q \quad \text{but} \quad \mathcal{N}, a \not\models p \triangleright q.$$

All propositional variables other than  $p, q$  are interpreted as  $\emptyset$  in both models.

**Model  $\mathcal{M}$ .** Let

$$W_M = \{w, u\},$$

with  $w \leq_M u$  and reflexive pairs. Let

$$R_M = \{(w, w), (w, u), (u, u)\}.$$

Let

$$V_M(p) = \{w, u\}, \quad V_M(q) = \{u\}.$$

Both valuations are upward closed. Then

$$S_M = R_M \cap \leq_M = \{(w, w), (w, u), (u, u)\}.$$

The model is based on an  $S$ -serial bare frame. Bare-frame forward confluence is immediate except for the refinements  $w \leq_M u$ , where  $u$  itself witnesses the required lift. Moreover,

$$\mathcal{M}, w \models p \triangleright q.$$

Indeed, at  $v = w$  choose the  $S_M$ -successor  $u$ , and at  $v = u$  choose the  $S_M$ -successor  $u$  again.

**Model  $\mathcal{N}$ .** Let

$$W_N = \{a, b, c, d\}.$$

Let  $\leq_N$  be the reflexive-transitive closure of

$$\{(a, b), (b, d), (c, d)\}.$$

Thus, in addition to reflexive pairs, we have

$$a \leq_N b, \quad b \leq_N d, \quad c \leq_N d, \quad a \leq_N d.$$

Let

$$R_N = \{(a, a), (a, b), (a, c), (b, b), (b, c), (b, d), (c, c), (d, d)\}.$$

Let

$$V_N(p) = W_N, \quad V_N(q) = \{c, d\}.$$

Both valuations are upward closed; for  $V_N(q)$ , this uses  $c \leq_N d$  and the fact that  $d$  has no proper  $\leq_N$ -successor. Then

$$S_N = R_N \cap \leq_N = \{(a, a), (a, b), (b, b), (b, d), (c, c), (d, d)\}.$$

The frame is  $S$ -serial, witnessed by the displayed  $S_N$ -loops and the edges  $aS_N b$  and  $bS_N d$ . It is bare. The nontrivial (FC) checks are:

| $x \leq x'$ | $xR_N u$ | witness $u'$ | check              |
|-------------|----------|--------------|--------------------|
| $a \leq b$  | $aR_N a$ | $b$          | $bR_N b, a \leq b$ |
| $a \leq b$  | $aR_N b$ | $d$          | $bR_N d, b \leq d$ |
| $a \leq b$  | $aR_N c$ | $c$          | $bR_N c, c \leq c$ |
| $a \leq d$  | $aR_N a$ | $d$          | $dR_N d, a \leq d$ |
| $a \leq d$  | $aR_N b$ | $d$          | $dR_N d, b \leq d$ |
| $a \leq d$  | $aR_N c$ | $d$          | $dR_N d, c \leq d$ |
| $b \leq d$  | $bR_N b$ | $d$          | $dR_N d, b \leq d$ |
| $b \leq d$  | $bR_N c$ | $d$          | $dR_N d, c \leq d$ |
| $b \leq d$  | $bR_N d$ | $d$          | $dR_N d, d \leq d$ |
| $c \leq d$  | $cR_N c$ | $d$          | $dR_N d, c \leq d$ |

At  $a$ , the formula  $p \triangleright q$  fails. Indeed, for  $v = a$ , the  $S_N$ -successors are  $a$  and  $b$ , and neither is in  $V_N(q)$ . Hence

$$\mathcal{N}, a \not\models p \triangleright q.$$

Now define

$$Z = \{(w, a), (w, b), (u, c), (u, d)\}.$$

We verify that  $Z$  is an IK-bisimulation.

Atomic harmony holds:  $p$  is true everywhere in both models;  $q$  is false at  $w, a, b$  and true at  $u, c, d$ ; and all other variables are false everywhere.

For the  $\leq$ -conditions, the only nontrivial checks are:

$$\begin{aligned} \text{Zig- } \leq: & \quad (w, a), w \leq_M u \quad a \leq_N d, (u, d) \in Z, \\ & \quad (w, b), w \leq_M u \quad b \leq_N d, (u, d) \in Z; \\ \text{Zag- } \leq: & \quad (w, a), a \leq_N b \quad w \leq_M w, (w, b) \in Z, \\ & \quad (w, a), a \leq_N d \quad w \leq_M u, (u, d) \in Z, \\ & \quad (w, b), b \leq_N d \quad w \leq_M u, (u, d) \in Z, \\ & \quad (u, c), c \leq_N d \quad u \leq_M u, (u, d) \in Z. \end{aligned}$$

Reflexive cases are immediate.

For the  $R$ -conditions, the nontrivial checks are:

$$\begin{aligned}
\text{Zig-}R : \quad & (w, a), wR_M w \quad aR_N a, (w, a) \in Z, \\
& (w, a), wR_M u \quad aR_N c, (u, c) \in Z, \\
& (w, b), wR_M w \quad bR_N b, (w, b) \in Z, \\
& (w, b), wR_M u \quad bR_N c, (u, c) \in Z, \\
& (u, c), uR_M u \quad cR_N c, (u, c) \in Z, \\
& (u, d), uR_M u \quad dR_N d, (u, d) \in Z; \\
\text{Zag-}R : \quad & (w, a), aR_N a \quad wR_M w, (w, a) \in Z, \\
& (w, a), aR_N b \quad wR_M w, (w, b) \in Z, \\
& (w, a), aR_N c \quad wR_M u, (u, c) \in Z, \\
& (w, b), bR_N b \quad wR_M w, (w, b) \in Z, \\
& (w, b), bR_N c \quad wR_M u, (u, c) \in Z, \\
& (w, b), bR_N d \quad wR_M u, (u, d) \in Z, \\
& (u, c), cR_N c \quad uR_M u, (u, c) \in Z, \\
& (u, d), dR_N d \quad uR_M u, (u, d) \in Z.
\end{aligned}$$

Thus  $Z$  is an IK-bisimulation and  $(w, a) \in Z$ . By Theorem 4.10, every  $\mathcal{L}_{\text{IK}}$ -formula has the same truth value at  $\mathcal{M}, w$  and  $\mathcal{N}, a$ . Since  $p \triangleright q$  separates these two pointed models, no formula of  $\mathcal{L}_{\text{IK}}$  can be equivalent to  $p \triangleright q$  over  $S$ -serial bare frames.  $\square$

*Remark 4.12* (Why IK cannot see conduction). The irreducibility result is not an accident of the chosen example. The IK fragment sees the vertical preorder  $\leq$  through intuitionistic implication and sees the horizontal relation  $R$  through  $\Box$  and  $\Diamond$ . What it cannot test is whether a particular  $R$ -edge is simultaneously a  $\leq$ -edge from the same source. Conduction is exactly sensitive to that mixed relation

$$S = R \cap \leq.$$

In the language of **PRel**, IK observes the persistent horizontal endomorphism  $R$ , but it cannot define the ambient subrelation  $S$  selected by the vertical order.

**Theorem 4.13** (Pointwise truth conservativity for IK reducts). *Let  $\mathcal{M} = \langle W, \leq, R, V \rangle$  be any model based on a bare frame. For every  $\varphi \in \mathcal{L}_{\text{IK}}$  and every  $w \in W$ ,*

$$\mathcal{M}, w \models \varphi \iff \mathcal{M}^{\text{IK}}, w \models \varphi.$$

*Proof.* By induction on  $\varphi \in \mathcal{L}_{\text{IK}}$ . The satisfaction clauses for  $\mathcal{L}_{\text{IK}}$ -formulas are exactly the IK clauses in both  $\mathcal{M}$  and  $\mathcal{M}^{\text{IK}}$ ; none of them mentions  $\triangleright$ . No  $S$ -seriality assumption is needed.  $\square$

*Remark 4.14* (On proof-theoretic conservativity). Theorem 4.13 is a pointwise truth-conservativity statement over a fixed model and its IK-reduct. It does not assert proof-theoretic conservativity over bare IK. Once the Hilbert system **H-DWS** is introduced, it will derive the IK-formula  $\alpha \rightarrow \Diamond \alpha$  (Lemma 5.6(v)), which is valid in  $S$ -serial bare frames but not valid over arbitrary bare IK frames. Thus the natural base for proof-theoretic comparison is

$$\mathbf{IK}_S = \mathbf{IK} + (\alpha \rightarrow \Diamond \alpha),$$

introduced in Definition 5.1, rather than bare IK.

## 5 The Hilbert System H-DWS

This section presents a Hilbert system for DWS. The guiding idea is that the proof theory should mirror the semantic position of conduction established in §4: conduction sits between a

necessary implication and an implication into the ordinary diamond, and it is the coreflected residual of the raw powerset existential along the ambient relation  $S = R \cap \leq$ .

As above,  $\top$  abbreviates  $\perp \rightarrow \perp$ , and  $\neg\alpha$  abbreviates  $\alpha \rightarrow \perp$ . Unless explicitly restricted to  $\mathcal{L}_{\text{IK}}$ , schematic variables range over formulas of  $\mathcal{L}_{\text{DWS}}$ .

**Definition 5.1** ( $\mathbf{IK}_S$ ). Let

$$\mathbf{IK}_S = \mathbf{IK} + (\alpha \rightarrow \Diamond\alpha),$$

where  $\alpha \rightarrow \Diamond\alpha$  is an axiom schema of the IK-language  $\mathcal{L}_{\text{IK}}$ . The subscript  $S$  anticipates Proposition 5.2, which shows that this schema characterises  $S$ -seriality of bare frames.

**Proposition 5.2** (Frame characterisation of  $\mathbf{IK}_S$ ). *A bare frame validates the schema  $\alpha \rightarrow \Diamond\alpha$ , for all IK-substitutions, if and only if it is  $S$ -serial. The same argument applies to  $\mathcal{L}_{\text{DWS}}$ -substitutions, using Theorem 2.3.*

*Proof. Soundness.* Suppose  $\mathcal{F}$  is  $S$ -serial. Let  $\mathcal{M}$  be any model on  $\mathcal{F}$ , and let  $w \in W$ . To show  $w \models \alpha \rightarrow \Diamond\alpha$ , let  $v \geq w$  and suppose  $v \models \alpha$ . By (Ser), choose  $u$  with  $vSu$ . Then  $vRu$  and  $v \leq u$ . By persistence,  $u \models \alpha$ . Hence  $v \models \Diamond\alpha$ . Therefore  $w \models \alpha \rightarrow \Diamond\alpha$ .

*Frame completeness.* Suppose  $\mathcal{F}$  is not  $S$ -serial. Then some world  $v_0$  has no  $S$ -successor. Define an upward-closed valuation by

$$V(p) = \uparrow v_0 = \{x \in W \mid v_0 \leq x\}.$$

Then  $v_0 \models p$ . But  $v_0 \not\models \Diamond p$ : if  $v_0Ru$  and  $u \models p$ , then  $u \in \uparrow v_0$ , so  $v_0 \leq u$ , and hence  $v_0Su$ , contradicting the choice of  $v_0$ . Thus  $v_0 \not\models p \rightarrow \Diamond p$ , witnessed by the extension  $v = v_0$ . Hence the schema is not valid on  $\mathcal{F}$ .  $\square$

*Remark 5.3* (Why  $\Diamond\top$  is not enough). The formula  $\Diamond\top$  characterises ordinary  $R$ -seriality,

$$\forall w \exists u (wRu),$$

not  $S$ -seriality. Forward confluence does not bridge this gap. For example, let

$$W = \{w, u\},$$

let  $\leq$  be the identity preorder, and let

$$R = \{(w, u), (u, u)\}.$$

This is an  $R$ -serial bare frame, since  $\leq$  is the identity and (FC) is immediate. But

$$S = R \cap \leq = \{(u, u)\},$$

so  $w$  has no  $S$ -successor. The formula  $\Diamond\top$  is valid, but the schema  $\alpha \rightarrow \Diamond\alpha$  fails: take  $V(p) = \{w\}$ . Then  $w \models p$ , while  $w \not\models \Diamond p$ .

We also do not replace  $\Diamond\top$  by  $\neg\Box\perp$ . With the stabilised box-clause used here,

$$w \models \neg\Box\perp \quad \text{iff} \quad \forall v \geq w \exists x \geq v \exists u (xRu),$$

which is a cofinal form of  $R$ -seriality above  $w$ , not pointwise  $R$ -seriality at  $w$ , and still not  $S$ -seriality.

**Definition 5.4** (The system  $\mathbf{H-DWS}$ ). The Hilbert system  $\mathbf{H-DWS}$  is formulated in the language  $\mathcal{L}_{\text{DWS}}$ . It contains all axiom schemata of IK, with schematic variables ranging over  $\mathcal{L}_{\text{DWS}}$ -formulas, the seriality schema

$$\alpha \rightarrow \Diamond\alpha,$$

and the following conduction schemata:

|                                                                                                                     |                                   |      |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------|------|
| $\Box(\varphi \rightarrow \psi) \rightarrow (\varphi \triangleright \psi)$                                          | <i>(necessity bound)</i>          | (C1) |
| $(\varphi \triangleright \psi) \rightarrow (\varphi \rightarrow \Diamond \psi)$                                     | <i>(possibility bound)</i>        | (C2) |
| $(\varphi \triangleright \psi) \wedge \Box(\psi \rightarrow \chi) \rightarrow (\varphi \triangleright \chi)$        | <i>(consequent strengthening)</i> | (C3) |
| $(\varphi \triangleright \psi) \wedge \alpha \rightarrow (\varphi \triangleright (\psi \wedge \alpha))$             | <i>(ambient persistence)</i>      | (CP) |
| $(\alpha \rightarrow \varphi) \rightarrow ((\varphi \triangleright \psi) \rightarrow (\alpha \triangleright \psi))$ | <i>(antecedent weakening)</i>     | (CW) |
| $(\varphi \triangleright \psi) \rightarrow (\varphi \triangleright (\psi \wedge \varphi))$                          | <i>(source persistence)</i>       | (SP) |

The inference rules are modus ponens, necessitation ( $\varphi/\Box\varphi$ ), and uniform substitution. Necessitation applies to theorems of the full  $\mathcal{L}_{\text{DWS}}$ -system.

*Remark 5.5* (Operator reading of the conduction axioms). For persistent predicates  $A, B, C$ , the first two axioms are exactly the proof-theoretic version of the modal sandwich

$$\Box_R(A \Rightarrow B) \subseteq A \triangleright_S B \subseteq A \Rightarrow E_R(B)$$

from Proposition 4.1. The remaining axioms express structural properties of conduction as a coreflected residual:

$$\begin{aligned} (A \triangleright_S B) \cap \Box_R(B \Rightarrow C) &\subseteq A \triangleright_S C, \\ (A \triangleright_S B) \cap C &\subseteq A \triangleright_S (B \cap C), \\ (C \Rightarrow A) &\subseteq ((A \triangleright_S B) \Rightarrow (C \triangleright_S B)), \end{aligned}$$

and

$$A \triangleright_S B \subseteq A \triangleright_S (B \cap A).$$

The axiom CP transports an ambient persistent fact  $\alpha$  from the source world through the chain  $w \leq v \leq u$ . The axiom SP transports the triggering antecedent  $\varphi$  from the world  $v$  at which conduction is applied to its  $S$ -successor  $u$ , using  $S \subseteq \leq$ . Thus CP and SP encode distinct persistence mechanisms.

**Lemma 5.6** (Derived principles). *In  $\mathbf{H-DWS}$ :*

- (i)  $\vdash \varphi \triangleright \top$ ;
- (ii)  $\vdash \psi \rightarrow (\varphi \triangleright \psi)$ ;
- (iii)  $\vdash \neg\varphi \rightarrow (\varphi \triangleright \psi)$ ;
- (iv)  $\vdash \varphi \triangleright \varphi$ ;
- (v)  $\vdash \alpha \rightarrow \Diamond\alpha$ .

*Proof.* (i) IK proves  $\varphi \rightarrow \top$ . By necessitation,  $\vdash \Box(\varphi \rightarrow \top)$ . Applying C1 gives  $\vdash \varphi \triangleright \top$ .

(ii) From (i),  $\vdash \varphi \triangleright \top$ . By intuitionistic propositional reasoning,

$$\vdash \psi \rightarrow ((\varphi \triangleright \top) \wedge \psi).$$

Instantiate CP with the consequent  $\top$  and ambient formula  $\psi$ . We obtain

$$\vdash \psi \rightarrow (\varphi \triangleright (\top \wedge \psi)).$$

Next, IK proves  $(\top \wedge \psi) \rightarrow \psi$ , so necessitation gives  $\vdash \Box((\top \wedge \psi) \rightarrow \psi)$ . By C3 and propositional reasoning,

$$\vdash (\varphi \triangleright (\top \wedge \psi)) \rightarrow (\varphi \triangleright \psi).$$

Combining the two implications gives  $\vdash \psi \rightarrow (\varphi \triangleright \psi)$ .

(iii) IK proves  $\perp \rightarrow \psi$ , hence necessitation and C1 give  $\vdash \perp \triangleright \psi$ . Instantiate CW with antecedent  $\varphi$  and middle formula  $\perp$ :

$$\vdash (\varphi \rightarrow \perp) \rightarrow ((\perp \triangleright \psi) \rightarrow (\varphi \triangleright \psi)).$$

Using  $\vdash \perp \triangleright \psi$ , we obtain  $\vdash \neg\varphi \rightarrow (\varphi \triangleright \psi)$ .

(iv) By (i),  $\vdash \varphi \triangleright \top$ . Applying SP,

$$\vdash \varphi \triangleright (\top \wedge \varphi).$$

IK proves  $(\top \wedge \varphi) \rightarrow \varphi$ , so by necessitation and C3,

$$\vdash (\varphi \triangleright (\top \wedge \varphi)) \rightarrow (\varphi \triangleright \varphi).$$

Hence  $\vdash \varphi \triangleright \varphi$ .

(v) Apply C2 to (iv). Thus  $\vdash \alpha \rightarrow \Diamond\alpha$ . This is already included as a seriality axiom in Definition 5.4; the derivation shows that the seriality schema is redundant once the conduction axioms are present. We retain it in the definition to make the relationship with  $\mathbf{IK}_S$  transparent.  $\square$

**Corollary 5.7** (Redundancy of the seriality axiom). *The system obtained from  $\mathbf{H-DWS}$  by deleting the seriality schema  $\alpha \rightarrow \Diamond\alpha$  proves exactly the same theorems as  $\mathbf{H-DWS}$ .*

*Proof.* The deleted schema is derivable by Lemma 5.6(v), using only the conduction axioms and the IK rules.  $\square$

**Theorem 5.8** (Soundness of  $\mathbf{H-DWS}$ ). *Every theorem of  $\mathbf{H-DWS}$  is valid in all  $S$ -serial bare models.*

*Proof.* The IK axiom schemata are valid on all bare frames. The seriality schema  $\alpha \rightarrow \Diamond\alpha$  is valid on  $S$ -serial bare frames by Proposition 5.2. It remains to verify the conduction axioms.

C1. Let  $w$  be arbitrary. To verify  $\Box(\varphi \rightarrow \psi) \rightarrow (\varphi \triangleright \psi)$  at  $w$ , let  $x \geq w$  and suppose  $x \models \Box(\varphi \rightarrow \psi)$ . We show  $x \models \varphi \triangleright \psi$ . Let  $v \geq x$  and suppose  $v \models \varphi$ . By (Ser), choose  $u$  such that  $vSu$ . Then  $vRu$  and  $v \leq u$ . By persistence,  $u \models \varphi$ . Since  $x \models \Box(\varphi \rightarrow \psi)$ ,  $v \geq x$ , and  $vRu$ , we have  $u \models \varphi \rightarrow \psi$ . Hence  $u \models \psi$ . Therefore  $x \models \varphi \triangleright \psi$ .

C2. Let  $x \geq w$  and suppose  $x \models \varphi \triangleright \psi$ . We show  $x \models \varphi \rightarrow \Diamond\psi$ . Let  $v \geq x$  and suppose  $v \models \varphi$ . By conduction, choose  $u$  such that  $vSu$  and  $u \models \psi$ . Since  $S \subseteq R$ ,  $vRu$ , so  $v \models \Diamond\psi$ .

C3. Let  $x \geq w$  and suppose

$$x \models (\varphi \triangleright \psi) \wedge \Box(\psi \rightarrow \chi).$$

We show  $x \models \varphi \triangleright \chi$ . Let  $v \geq x$  and suppose  $v \models \varphi$ . From  $x \models \varphi \triangleright \psi$ , choose  $u$  such that  $vSu$  and  $u \models \psi$ . Since  $vRu$ ,  $v \geq x$ , and  $x \models \Box(\psi \rightarrow \chi)$ , we have  $u \models \psi \rightarrow \chi$ . Hence  $u \models \chi$ .

CP. Let  $x \geq w$  and suppose

$$x \models (\varphi \triangleright \psi) \wedge \alpha.$$

We show  $x \models \varphi \triangleright (\psi \wedge \alpha)$ . Let  $v \geq x$  and suppose  $v \models \varphi$ . From  $x \models \varphi \triangleright \psi$ , choose  $u$  such that  $vSu$  and  $u \models \psi$ . Since  $x \leq v \leq u$ , persistence gives  $u \models \alpha$ . Therefore  $u \models \psi \wedge \alpha$ .

CW. Let  $x \geq w$  and suppose  $x \models \alpha \rightarrow \varphi$ . We show

$$x \models (\varphi \triangleright \psi) \rightarrow (\alpha \triangleright \psi).$$

Let  $y \geq x$  and suppose  $y \models \varphi \triangleright \psi$ . We show  $y \models \alpha \triangleright \psi$ . Let  $v \geq y$  and suppose  $v \models \alpha$ . Since  $v \geq x$  and  $x \models \alpha \rightarrow \varphi$ , we have  $v \models \varphi$ . From  $y \models \varphi \triangleright \psi$ , choose  $u$  such that  $vSu$  and  $u \models \psi$ . Hence  $y \models \alpha \triangleright \psi$ .

*SP.* Let  $x \geq w$  and suppose  $x \models \varphi \triangleright \psi$ . We show  $x \models \varphi \triangleright (\psi \wedge \varphi)$ . Let  $v \geq x$  and suppose  $v \models \varphi$ . From  $x \models \varphi \triangleright \psi$ , choose  $u$  such that  $vSu$  and  $u \models \psi$ . Since  $vSu$  implies  $v \leq u$ , persistence gives  $u \models \varphi$ . Hence  $u \models \psi \wedge \varphi$ .

Modus ponens is sound by the definition of implication. Necessitation is sound because, if  $\theta$  is valid in all  $S$ -serial bare models, then every  $R$ -successor of every refinement satisfies  $\theta$ , so  $\Box\theta$  is valid. Uniform substitution is sound by the compositionality of the semantics and the upward-closure of formula extensions.  $\square$

## 6 The Labelled Sequent System LS-DWS

We extend the fully labelled sequent calculus of Marin, Morales, and Straßburger [5]. A *labelled sequent* has the form  $\mathcal{R}, \Gamma \Rightarrow \Delta$ , where  $\mathcal{R}$  is a finite set of *relational atoms* of the forms  $x \leq y$  and  $xRy$  (with  $x, y$  drawn from a countable set of labels),  $\Gamma$  is a finite multiset of labelled formulas  $x:\varphi$  placed on the *left* (asserting truth at  $x$ ), and  $\Delta$  is a finite multiset of labelled formulas placed on the *right* (asserting falsity at  $x$ ). We write  $xSy$  as shorthand for  $xRy$  and  $x \leq y$  (both present in  $\mathcal{R}$ ).

*Relationship to [5].* We inherit the structural rules (REFL, TRANS, FC) and the standard propositional and modal rules ( $\wedge, \vee, \rightarrow, \Box, \Diamond$ ) from [5], together with their soundness proofs. We modify three aspects: (i) the MON rule is strengthened to propagate *all* left-side labelled formulas (not only atoms) through  $\leq$ -edges; (ii) the SER rule generates both  $xRy$  and  $x \leq y$  for the fresh label  $y$ , producing an  $S$ -successor rather than a mere  $R$ -successor; (iii) we add annotations, the  $\triangleright$ -rules, and the redirect-based proof-search discipline. The completeness and termination arguments are entirely new and do not rely on the corresponding results in [5].

### 6.1 Challenger annotations

**Definition 6.1** (Annotated sequent). An *annotated sequent* is  $(\mathcal{R}, \mathcal{A}, \Gamma \Rightarrow \Delta)$  where  $\mathcal{A}$  is a set of annotations  $y^{(x, \varphi \triangleright \psi)}$ , recording that  $y$  is the challenger for  $x:\varphi \triangleright \psi \in \Delta$ .<sup>3</sup>

**Definition 6.2** (Annotation semantics). Under assignment  $\sigma$ , annotation  $y^{(x, \varphi \triangleright \psi)}$  is *satisfied* if  $\sigma(y) \models \varphi$  and  $\forall u (\sigma(y) Su \Rightarrow u \not\models \psi)$ . Intuitively, the annotation records *one* witness to the falsification of  $\sigma(x) \models \varphi \triangleright \psi$ : the world  $\sigma(y)$  is the existentially quantified  $v \geq \sigma(x)$  at which  $\varphi$  holds and no  $S$ -successor satisfies  $\psi$ . An annotated sequent is *valid* if every assignment satisfying all relational atoms, annotations, and  $\Gamma$  also satisfies some formula in  $\Delta$ .

### 6.2 Rules

Structural rules (REFL, TRANS, MON, FC, SER) and propositional/modal rules ( $\wedge, \vee, \rightarrow, \Box, \Diamond$ ) are as in [5], with two modifications: the MON rule propagates *all* left-side labelled formulas (not only atoms) through  $\leq$ -edges, and SER produces both  $xRy$  and  $x \leq y$  for fresh  $y$ .

The conduction rules:

**Left rule** (branching):

$$\frac{\mathcal{R}, x \leq y, x:\varphi \triangleright \psi, \Gamma \Rightarrow \Delta, y:\varphi \mid \mathcal{R}, x \leq y, x:\varphi \triangleright \psi, yRz, y \leq z, z:\psi, \Gamma \Rightarrow \Delta}{\mathcal{R}, x \leq y, x:\varphi \triangleright \psi, \Gamma \Rightarrow \Delta} \triangleright L$$

( $z$  fresh in right premise.)

<sup>3</sup>Annotation mechanisms of this general kind—where labels carry auxiliary indices recording semantic obligations—have been used in labelled calculi for conditional logics and logics of agency to maintain the subformula property and ensure rule-level soundness; see, e.g., the indexed-world mechanisms in Negri’s labelled calculi for conditional logics. Our annotated-challenger device fits into this broader tradition of annotated labelled deduction.

**Right rules:**

$$\frac{\mathcal{R}, x \leq y, \mathcal{A} \cup \{y^{(x, \varphi \triangleright \psi)}\}, y: \varphi, \Gamma \Rightarrow \Delta, x: \varphi \triangleright \psi}{\mathcal{R}, \mathcal{A}, \Gamma \Rightarrow \Delta, x: \varphi \triangleright \psi} \triangleright R_1$$

( $y$  fresh.)

$$\frac{\mathcal{R}, x \leq y, yRz, y \leq z, \mathcal{A}, y: \varphi, \Gamma \Rightarrow \Delta, z: \psi, x: \varphi \triangleright \psi}{\mathcal{R}, x \leq y, yRz, y \leq z, \mathcal{A}, y: \varphi, \Gamma \Rightarrow \Delta, x: \varphi \triangleright \psi} \triangleright R_2$$

Side condition:  $y^{(x, \varphi \triangleright \psi)} \in \mathcal{A}$ .

**Lemma 6.3** (Isolation soundness of  $\triangleright R_2$ ).  $\triangleright R_2$  preserves validity of annotated sequents.

*Proof.* If the conclusion is falsified by  $\sigma$ , then  $\sigma(y) S \sigma(z)$  and the annotation gives  $\sigma(z) \not\models \psi$ , so  $z: \psi$  in  $\Delta$  is also falsified.  $\square$

**Theorem 6.4** (Soundness). *Every rule of LS-DWS preserves validity. Every provable formula is valid in all S-serial bare models.*

*Proof.* Structural and standard logical rules: as in [5].  $\triangleright L$ : standard branching argument via  $(S \triangleright)$ .  $\triangleright R_1$ : if the conclusion is falsified,  $\sigma(x) \not\models \varphi \triangleright \psi$  gives a witness  $v$ ; setting  $\sigma(y) = v$  satisfies the annotation.  $\triangleright R_2$ : Lemma 6.3.  $\square$

*Remark 6.5* (Motivation for annotations). Without annotations,  $\triangleright R_2$  is not validity-preserving. The counterexample:  $W = \{a, b, c\}$ ,  $\leq$  chain  $a \leq b \leq c$ ,  $R = \{(a, a), (b, c), (c, c)\}$ ,  $V(\varphi) = W$ ,  $V(\psi) = \{b, c\}$ . Then  $a \not\models \varphi \triangleright \psi$ , but with  $\sigma(x) = a, \sigma(y) = b, \sigma(z) = c$ , the unannotated conclusion is falsified while the unannotated premise is valid. The root cause is the  $\exists \forall$  structure of the  $\triangleright$ -falsification clause; the annotation restores the link between challenger and propagation target.

### 6.3 Completeness

**Definition 6.6** (Saturation). A branch  $\mathcal{B} = (\mathcal{R}, \mathcal{A}, \Gamma \Rightarrow \Delta)$  is *saturated* if not closed and no rule extends it. The conditions are:

(Sat-id): No  $x: \alpha \in \Gamma \cap \Delta$ ;  $x: \perp \notin \Gamma$ .

(Sat-refl):  $x \leq x \in \mathcal{R}$  for all  $x$ .

(Sat-trans):  $x \leq y, y \leq z \in \mathcal{R} \Rightarrow x \leq z \in \mathcal{R}$ .

(Sat-mon):  $x \leq y \in \mathcal{R}, x: \psi \in \Gamma \Rightarrow y: \psi \in \Gamma$ , for all  $\psi \in \text{Sub}(\varphi_0)$ .

(Sat-fc):  $x \leq y, xRz \in \mathcal{R} \Rightarrow \exists u (yRu, z \leq u \in \mathcal{R})$ .

(Sat-ser):  $\forall x \exists y (xRy, x \leq y \in \mathcal{R})$ .

(Sat- $\wedge L$ ):  $x: \alpha \wedge \beta \in \Gamma \Rightarrow x: \alpha, x: \beta \in \Gamma$ .

(Sat- $\wedge R$ ):  $x: \alpha \wedge \beta \in \Delta \Rightarrow x: \alpha \in \Delta$  or  $x: \beta \in \Delta$ .

(Sat- $\vee L$ ):  $x: \alpha \vee \beta \in \Gamma \Rightarrow x: \alpha \in \Gamma$  or  $x: \beta \in \Gamma$ .

(Sat- $\vee R$ ):  $x: \alpha \vee \beta \in \Delta \Rightarrow x: \alpha, x: \beta \in \Delta$ .

(Sat- $\rightarrow L$ ):  $x: \alpha \rightarrow \beta \in \Gamma, x \leq y \in \mathcal{R} \Rightarrow y: \alpha \in \Delta$  or  $y: \beta \in \Gamma$ .

(Sat- $\rightarrow R$ ):  $x: \alpha \rightarrow \beta \in \Delta \Rightarrow \exists y (x \leq y \in \mathcal{R}, y: \alpha \in \Gamma, y: \beta \in \Delta)$ .

(Sat- $\Box L$ ):  $x: \Box \alpha \in \Gamma, xRy \in \mathcal{R} \Rightarrow y: \alpha \in \Gamma$ .

(Sat- $\Box R$ ):  $x: \Box \alpha \in \Delta \Rightarrow \exists y (xRy \in \mathcal{R}, y: \alpha \in \Delta)$ .

(Sat- $\Diamond L$ ):  $x: \Diamond \alpha \in \Gamma \Rightarrow \exists y (xRy \in \mathcal{R}, y: \alpha \in \Gamma)$ .

(Sat- $\Diamond R$ ):  $x: \Diamond \alpha \in \Delta, xRy \in \mathcal{R} \Rightarrow y: \alpha \in \Delta$ .

(Sat- $\triangleright L$ ):  $x: \varphi \triangleright \psi \in \Gamma, x \leq y \in \mathcal{R} \Rightarrow y: \varphi \in \Delta$  or  $\exists z (yRz, y \leq z \in \mathcal{R}, z: \psi \in \Gamma)$ .

(Sat- $\triangleright R$ ):  $x: \varphi \triangleright \psi \in \Delta \Rightarrow \exists y (x \leq y \in \mathcal{R}, y^{(x, \varphi \triangleright \psi)} \in \mathcal{A}, y: \varphi \in \Gamma, \text{ and } \forall z (yRz, y \leq z \in \mathcal{R} \Rightarrow z: \psi \in \Delta))$ .

**Definition 6.7** (Extracted structure). Given a saturated branch  $\mathcal{B}$ , define  $\mathcal{M}_{\mathcal{B}} = \langle W, \leq, R, V \rangle$  by:  $W$  = set of labels;  $x \leq y$  iff  $x \leq y \in \mathcal{R}$ ;  $xRy$  iff  $xRy \in \mathcal{R}$ ;  $V(p) = \{x : x: p \in \Gamma\}$ .

**Lemma 6.8** (Truth Lemma). *For every  $\varphi \in \text{Sub}(\varphi_0)$  and label  $x$ :*

- (i)  $x:\varphi \in \Gamma \Rightarrow \mathcal{M}_{\mathcal{B}}, x \models \varphi$ .
- (ii)  $x:\varphi \in \Delta \Rightarrow \mathcal{M}_{\mathcal{B}}, x \not\models \varphi$ .

*Proof.* By induction on the complexity of  $\varphi$ . The IK cases use the corresponding saturation conditions exactly as in [5], *modulo* our strengthened MON rule: the standard MMS21 arguments are reused in an adapted saturation environment where MON propagates all left-side formulas (not only atoms) through  $\leq$ -edges. This strengthening only adds formulas that the MMS21 arguments already assume to be available at  $\leq$ -successors; it therefore does not invalidate any inherited case. The key observation is that (Sat-mon) propagates all left-side formulas through  $\leq$ , so the  $\Box$  case of (i) proceeds as follows.

Suppose  $x:\Box\alpha \in \Gamma$  and  $v \geq x$  with  $vRu$ . By (Sat-mon),  $v:\Box\alpha \in \Gamma$ . By (Sat- $\Box$ L),  $u:\alpha \in \Gamma$ . By IH(i),  $u \models \alpha$ .

*Case  $\varphi \triangleright \psi$ , part (i).* Suppose  $x:\varphi \triangleright \psi \in \Gamma$  and  $y \geq x$  with  $y \models \varphi$ . By (Sat-mon),  $y:\varphi \triangleright \psi \in \Gamma$ . By (Sat-refl),  $y \leq y \in \mathcal{R}$ . By (Sat- $\triangleright$ L) applied to  $y:\varphi \triangleright \psi \in \Gamma$  and  $y \leq y$ : either  $y:\varphi \in \Delta$  (contradicting  $y \models \varphi$  by IH(ii)) or  $\exists z (ySz, z:\psi \in \Gamma)$ . By IH(i):  $z \models \psi$ .

*Case  $\varphi \triangleright \psi$ , part (ii).* Suppose  $x:\varphi \triangleright \psi \in \Delta$ . By (Sat- $\triangleright$ R), there exists a challenger  $y$  with  $x \leq y \in \mathcal{R}$ ,  $y^{(x, \varphi \triangleright \psi)} \in \mathcal{A}$ ,  $y:\varphi \in \Gamma$ , and for every  $z$  with  $yRz, y \leq z \in \mathcal{R}$ :  $z:\psi \in \Delta$ . By IH(i):  $y \models \varphi$ . By IH(ii): for every  $S$ -successor  $z$  of  $y$  in  $\mathcal{M}_{\mathcal{B}}$ ,  $z \not\models \psi$ . The world  $y$  therefore witnesses the falsification of the  $\triangleright$ -clause:  $y \geq x$ ,  $y \models \varphi$ , and no  $S$ -successor of  $y$  satisfies  $\psi$ . Hence  $x \not\models \varphi \triangleright \psi$ .  $\square$

*Remark 6.9.* Because (Sat-mon) propagates all left-side formulas, the truth lemma proof never invokes forward confluence or seriality *directly in the inductive steps*. Persistence is handled locally by (Sat-mon), which supplies the needed formula at each  $\leq$ -successor without appealing to frame properties. The countermodel construction (Proposition 6.21), by contrast, *does* rely on the saturation conditions (Sat-fc) and (Sat-ser) holding at all labels, in order to guarantee that the extracted structure is a valid  $S$ -serial bare model. The distinction matters: the truth lemma's inductive argument is frame-property-free, but its *applicability* depends on every label being fully saturated.

## 6.4 Termination via the redirect mechanism

We now define the proof-search algorithm precisely and prove termination. The subsection is organised around four interdependent definitions—the algorithm (Definition 6.10), triggers (Definition 6.11), redirect (Definition 6.14), and branch state (Definition 6.19)—followed by the termination proof (Lemma 6.20), which ties them together.

The key idea is a *redirect mechanism*: when the algorithm would create a fresh label whose formula assignment (type) already appears at the same depth level, it redirects to the existing label instead. This ensures that every label carries all saturation conditions—including the P4 conditions needed for the truth lemma—without creating unbounded chains.

**Definition 6.10** (Proof-search algorithm). Given input  $\varphi_0$ , let  $\Sigma = \text{Sub}(\varphi_0)$ ,  $n = |\Sigma|$ ,  $d = \text{md}(\varphi_0) \leq n$ . The algorithm maintains a branch state and applies rules in priority order:

- P1. *Closure*: if  $x:\alpha \in \Gamma \cap \Delta$  or  $x:\perp \in \Gamma$ , close.
- P2. *Propagation* (non-generating): REFL, TRANS, MON,  $\wedge$ L,  $\vee$ R,  $\Box$ L,  $\Diamond$ R,  $\triangleright$ R<sub>2</sub>.
- P3. *Branching*:  $\wedge$ R,  $\vee$ L,  $\rightarrow$ L,  $\triangleright$ L.
- P4. *Modal generation* (creating labels at lower depth):  $\Box$ R,  $\Diamond$ L,  $\triangleright$ L right premise. Subject to redirect (Definition 6.14).

P5. *Same-depth generation*:  $\rightarrow R$ ,  $\triangleright R_1$ , SER, FC. Subject to redirect (Definition 6.14).

Each rule is applied at most once per trigger (Definition 6.11). The priority ordering ensures *fairness*: at each step, the algorithm selects the highest-priority applicable rule instance whose trigger has not yet been processed. Since the trigger space is finite (Definition 6.11 and Lemma 6.18), every applicable rule instance is eventually applied. A branch is *saturated* when no unprocessed trigger remains; it is *closed* if P1 succeeds.

**Definition 6.11** (Trigger). A *trigger* is a tuple that uniquely identifies one potential rule application. For each rule class:

- **P1** (closure):  $(ID, x, \alpha)$  where  $x:\alpha \in \Gamma \cap \Delta$ , or  $(BOT, x)$  where  $x:\perp \in \Gamma$ .
- **P2** (propagation):  $(REFL, x)$ ,  $(TRANS, x \leq y, y \leq z)$ ,  $(MON, x \leq y, x:\alpha \in \Gamma)$ ,  $(\Box L, x:\Box \alpha \in \Gamma, xRy)$ ,  $(\Diamond R, x:\Diamond \alpha \in \Delta, xRy)$ ,  $(\triangleright R_2, y^{(x, \varphi \triangleright \psi)}, ySz)$ .
- **P3** (branching):  $(RULE, x, \varphi)$  encoding the principal label, the side, and the principal formula.
- **P4/P5** (generation):  $(RULE, x, \varphi)$  for rules triggered by a formula at  $x$ ;  $(SER, x)$  for seriality;  $(FC, x \leq y, xRz)$  for forward confluence.

The set of all possible triggers on a branch with labels  $\mathcal{L}$ , subformulas  $\Sigma$ , and relational atoms  $\mathcal{R}$  is denoted  $\text{Trig}(\mathcal{L}, \Sigma, \mathcal{R})$ . Its cardinality is bounded by  $O(N^3 + N^2n)$ , where  $N$  is the label bound.<sup>4</sup>

*Remark 6.12.* Both  $\rightarrow R$  and  $\triangleright R_1$  are classified as same-depth (P5), not depth-reducing (P4), because they create  $\leq$ -successors rather than  $R$ -successors. For  $\rightarrow R$ :  $\text{md}(\alpha \rightarrow \beta) = \max(\text{md}(\alpha), \text{md}(\beta))$ , with no modal-depth contribution from  $\rightarrow$ . For  $\triangleright R_1$ : the challenger  $y$  receives  $y:\varphi \in \Gamma$  with  $\text{md}(\varphi) \leq \text{md}(\varphi \triangleright \psi) - 1$ , but (Sat-mon) immediately imports all of  $x$ 's  $\Gamma$ -formulas at their original depth. Since the challenger is a  $\leq$ -successor of  $x$ , classifying it at the same depth as  $x$  correctly reflects the fact that its formula inventory is no smaller than that of  $x$ .

The *actual* depth decrease occurs at  $S$ -successor creation: the rules  $\Box R$ ,  $\Diamond L$ , and the right premise of  $\triangleright L$  (which creates an  $S$ -successor with  $yRz$  and  $y \leq z$ ) are classified as P4. Note that P4 rules may fire at any depth level, including depth 0: because the  $\leq$ -edge  $y \leq z$  created by a P4 rule triggers mon-propagation of all of  $y$ 's  $\Gamma$ -formulas to  $z$ , a depth-0 label may carry formulas of arbitrarily high modal depth. To ensure that label depths remain non-negative, P4 rules use the convention  $\delta(z) = \max(\delta(y) - 1, 0)$  (Definition 6.13). When a P4 rule fires at depth 0, it creates a depth-0 label, which the redirect mechanism handles via the type registry at depth 0 exactly as at any other level. This does not affect the label bound or termination.

**Definition 6.13** (Label depth). The root  $x_0$  has  $\delta(x_0) = d$ . If  $x$  creates  $y$  via a P4 rule ( $\Box R$ ,  $\Diamond L$ , or  $\triangleright L$  right premise), then  $\delta(y) = \max(\delta(x) - 1, 0)$ . If  $x$  creates  $y$  via a P5 rule ( $\rightarrow R$ ,  $\triangleright R_1$ , SER), then  $\delta(y) = \delta(x)$ . More precisely, FC applied to the pair  $(x \leq y, xRz)$  creates  $u$  with  $yRu$  and  $z \leq u$ ; the depth of  $u$  equals  $\delta(z)$ , since  $u$  extends the  $\leq$ -chain of  $z$ . All label depths are non-negative by construction.

**Definition 6.14** (Type and redirect). The *type* of label  $x$  is  $\tau(x): \Sigma \rightarrow \{L, R, \emptyset\}$ , recording membership in  $\Gamma$ ,  $\Delta$ , or neither.<sup>5</sup> There are at most  $3^n$  types.

<sup>4</sup>The  $N^3$  term accounts for triggers involving three labels, such as TRANS triggers  $(x \leq y, y \leq z)$  and FC triggers  $(x \leq y, xRz)$ . The  $N^2n$  term accounts for formula-driven triggers at pairs of labels. In particular,  $\triangleright R_2$  triggers have the form  $(y^{(x, \varphi \triangleright \psi)}, yRz)$ : since annotations number at most  $Nn$  and each determines the label  $y$ , the factor  $z$  ranges over  $N$  labels, giving at most  $N^2n$  such triggers.

<sup>5</sup>The codomain  $\{L, R, \emptyset\}$  suffices because closure (P1) has highest priority: any label with the same formula on both sides is closed before types are computed. Thus no formula is simultaneously in  $\Gamma$  and  $\Delta$  at type-computation time.

For each depth level  $\ell \in \{0, \dots, d\}$ , the algorithm maintains a *type registry*: a partial function  $\text{reg}_\ell: \{L, R, \emptyset\}^\Sigma \rightarrow \text{Labels}$  mapping each type to the first label created at depth  $\ell$  with that type.

When a rule would create a fresh label  $y$  at depth  $\ell$ :

1. Create  $y$  provisionally. Add the relational atoms and labelled formulas specified in the premise of the generating rule: for  $\Box R$  applied to  $x:\Box\alpha \in \Delta$ , the atom  $xRy$  and the formula  $y:\alpha \in \Delta$ ; for  $\Diamond L$  applied to  $x:\Diamond\alpha \in \Gamma$ , the atom  $xRy$  and the formula  $y:\alpha \in \Gamma$ ; for  $\triangleright R_1$ , the atom  $x \leq y$ , the annotation  $y^{(x, \varphi \triangleright \psi)}$ , and  $y:\varphi \in \Gamma$ ; for the right premise of  $\triangleright L$ , the atoms  $yRz$ ,  $y \leq z$  and  $z:\psi \in \Gamma$ ; for SER, the atoms  $xRy$  and  $x \leq y$ ; for FC, the atoms  $yRu$  and  $z \leq u$ .
2. Apply all P2 rules at  $y$  (propagation) until saturation, then read off  $\tau(y)$ . (The type is well-defined because P2 rules only add formulas and relational atoms; they do not remove them. Since the branch is finite, P2 saturation terminates, yielding a stable type for  $y$ .)
3. **Registry check.** If  $\text{reg}_\ell(\tau(y))$  is defined—say it equals an existing label  $z$ —then:
  - (a) Delete  $y$ .
  - (b) For every relational atom  $a X y$  (where  $X \in \{\leq, R\}$ ) that was added for  $y$ , replace it with  $a X z$ . Similarly replace  $y X a$  with  $z X a$ .
  - (c) For every labelled formula  $y:\psi$  that was added, check whether  $z:\psi$  is already present; if not, add it. (Since  $\tau(z) = \tau(y)$ , every subformula of  $\varphi_0$  already present at  $y$  is also present at  $z$  on the same side; only formulas arising from relational propagation—such as those generated by the new relational atoms in step 3b—may be genuinely new at  $z$ .)
  - (d) *Annotation transfer.* For every annotation  $y^{(x', \varphi' \triangleright \psi')} \in \mathcal{A}$ , replace it with  $z^{(x', \varphi' \triangleright \psi')}$ . This ensures that the side condition for  $\triangleright R_2$  is met at  $z$ 's  $S$ -successors.
  - (e) Resume rule application at  $z$  and its relational neighbours: first P2 (propagation of any newly added atoms or formulas, including  $\triangleright R_2$  checks for transferred annotations at  $z$ 's existing  $S$ -successors), then any newly enabled P5 triggers (SER, FC) arising from the new relational atoms at  $z$ . Each such application processes a fresh trigger from the trigger space.
4. If  $\text{reg}_\ell(\tau(y))$  is undefined, set  $\text{reg}_\ell(\tau(y)) := y$  and continue with all rules (P2–P5) at  $y$ .

**Example 6.15** (Redirect in action). Suppose  $\text{Sub}(\varphi_0) = \{p, q, p \triangleright q\}$  and at depth  $\ell = 1$  the registry already contains a label  $z$  with type  $\tau(z) = (p \mapsto L, q \mapsto R, p \triangleright q \mapsto R)$ . Now  $\Box R$  fires at a label  $x$  of depth  $\delta(x) = 2$ , creating a provisional label  $y$  at depth  $\delta(y) = \max(2 - 1, 0) = 1$  (Definition 6.13); after P2 propagation,  $y$  acquires the same type:  $\tau(y) = \tau(z)$ . The registry check fires:  $y$  is deleted, its relational atoms are rewired to  $z$  (step 3b), the annotation  $y^{(x, p \triangleright q)}$  becomes  $z^{(x, p \triangleright q)}$  (step 3d), and  $\triangleright R_2$  is resumed at  $z$ 's  $S$ -successors (step 3e). No new label is created; only the processed trigger and (possibly) new relational atoms at  $z$  are added to the branch state.

*Remark 6.16.* Four properties of the redirect mechanism deserve emphasis.

First, *no label is blocked or terminal*. Every label in the branch receives all rules—P2 through P5—and therefore satisfies all saturation conditions. The truth lemma (Lemma 6.8) holds at every label without special-case arguments.

Second, redirections may create  $\leq$ -cycles: if  $z \leq a$  was previously established and the redirection adds  $a \leq z$ , then  $a$  and  $z$  become  $\leq$ -equivalent. This is harmless for the preorder and for the truth lemma, which evaluates formulas pointwise.

Third, redirections may add new  $R$ -edges at existing labels, which can trigger further P2/P5 applications (which may themselves attempt to create new labels, subject to further redirections). Each such cascade terminates because each step processes a fresh trigger (Definition 6.11), adding it to the processed-trigger set  $\mathcal{P}$  (Definition 6.19); since the trigger space is finite, the cascade is well-founded.

Fourth, annotations are transferred during redirection (step 3d): when  $y$  is redirected to  $z$ , every annotation  $y^{(x', \varphi' \triangleright \psi')}$  becomes  $z^{(x', \varphi' \triangleright \psi')}$ , ensuring that  $\triangleright R_2$  can fire at  $z$ 's  $S$ -successors. Step 3e then applies the corresponding  $\triangleright R_2$  checks.

**Lemma 6.17** (Redirect adequacy). *Let  $y$  be a provisional label at depth  $\ell$  that is redirected to an existing label  $z$  with  $\tau(z) = \tau(y)$  at the same depth. Then the resulting branch state is adequate: every saturation obligation that would have been discharged at  $y$  is discharged at  $z$ , and no future obligation relevant to the truth lemma or countermodel construction is lost.*

*Proof.* We verify the three components of adequacy.

*Formula obligations.* Since  $\tau(z) = \tau(y)$ , for every  $\psi \in \text{Sub}(\varphi_0)$  the label  $z$  carries  $\psi$  on the same side (left, right, or neither) as  $y$  would. Every rule whose trigger depends only on formula content at  $y$  (i.e., the formula-based triggers of P2–P5, excluding the annotation-dependent  $\triangleright R_2$ ) has a corresponding instance at  $z$  with the same formulas.

*Relational obligations.* Step 3b of the redirect transfers all relational atoms from  $y$  to  $z$ . Step 3e resumes rule application at  $z$ : first P2 propagation rules (REFL, TRANS, MON), then any newly enabled P5 triggers (SER, FC) arising from the transferred relational atoms. Each such application processes a fresh trigger, and any further P4/P5 applications triggered by the new atoms terminate by redirect monotonicity (Lemma 6.20).

*Annotation obligations.* Step 3d transfers every annotation  $y^{(x', \varphi' \triangleright \psi')}$  to  $z^{(x', \varphi' \triangleright \psi')}$ . These transferred annotations may create *new*  $\triangleright R_2$  triggers at  $z$  that were not determined by formula-type alone: Definition 6.11 includes triggers of the form  $(\triangleright R_2, z^{(x', \varphi' \triangleright \psi')}, zSw)$  for each existing  $S$ -successor  $w$  of  $z$ . Step 3e explicitly processes these annotation-dependent triggers, applying the  $\triangleright R_2$  check at each  $S$ -successor of  $z$  for each transferred annotation. Thus every  $\psi'$ -placement that would have occurred at an  $S$ -successor of  $y$  is now attempted at the corresponding  $S$ -successor of  $z$ . Since the truth lemma evaluates formulas at each label using only the saturation conditions holding at that label (Remark 6.9), and  $z$  carries the same type and all transferred annotations with their obligations discharged by step 3e, no truth-lemma case is affected. Likewise, the countermodel construction (Proposition 6.21) depends only on the saturation conditions at active labels, all of which are preserved.  $\square$

**Lemma 6.18** (Label bound). *The total number of labels is at most  $N \leq (d+1) \cdot 3^n \leq (n+1) \cdot 3^n$ .*

*Proof.* At each depth level  $\ell$ , the type registry admits at most  $3^n$  entries (one per type). With  $d+1$  depth levels, the total number of labels is at most  $(d+1) \cdot 3^n$ . Every label that would duplicate a type at its depth level is redirected to the existing label; no new label is created.  $\square$

**Definition 6.19** (Branch state). The *branch state* is the tuple  $\mathcal{S} = (\mathcal{L}, \mathcal{R}, \Gamma, \Delta, \mathcal{A}, \mathcal{T}, \mathcal{P})$ , where:

- $\mathcal{L} \subseteq \text{Labels}$ : the set of active labels (at most  $N$ );
- $\mathcal{R} \subseteq \mathcal{L}^2 \times \{\leq, R\}$ : the set of relational atoms (at most  $2N^2$ );
- $\Gamma \subseteq \mathcal{L} \times \Sigma$ : left formula placements (at most  $Nn$ );
- $\Delta \subseteq \mathcal{L} \times \Sigma$ : right formula placements (at most  $Nn$ );
- $\mathcal{A} \subseteq \mathcal{L} \times \mathcal{L} \times \text{Sub}_{\triangleright}(\varphi_0)$ : the annotation set (at most  $Nn$ );
- $\mathcal{T}: \{0, \dots, d\} \rightarrow (\{L, R, \emptyset\}^\Sigma \rightarrow \mathcal{L})$ : the type registry;

- $\mathcal{P} \subseteq \text{Trig}(\mathcal{L}, \Sigma, \mathcal{R})$ : the set of *processed triggers* (Definition 6.11).

We order branch states by component-wise inclusion:  $\mathcal{S} \sqsubseteq \mathcal{S}'$  iff each component of  $\mathcal{S}$  is contained in (or, for  $\mathcal{T}$ , extended by) the corresponding component of  $\mathcal{S}'$ . In particular  $\mathcal{P} \subseteq \mathcal{P}'$ . Every rule application adds the processed trigger to  $\mathcal{P}$ , so  $\mathcal{P}$  grows monotonically regardless of whether any other component changes.

**Lemma 6.20** (Termination). *The proof-search procedure terminates on any input.*

*Proof.* By Lemma 6.18,  $|\mathcal{L}| \leq N = (d+1) \cdot 3^n$ . The material state components satisfy  $|\mathcal{R}| + |\Gamma| + |\Delta| + |\mathcal{A}| \leq 2N^2 + 3Nn$ . The trigger space  $\text{Trig}(\mathcal{L}, \Sigma, \mathcal{R})$  has cardinality at most  $T = O(N^3 + N^2n)$  (Definition 6.11).

*Strict progress via processed triggers.* Every rule application processes exactly one trigger and adds it to  $\mathcal{P}$ . Since a trigger is processed at most once (Definition 6.10),  $|\mathcal{P}|$  increases strictly at every step. Because  $|\mathcal{P}| \leq T$  and  $T$  is finite, the procedure terminates after at most  $T$  steps per branch.

*Monotonicity of the material state by priority class.* In addition to the growth of  $\mathcal{P}$ , we verify that the material components behave monotonically:

- **P1** (closure): terminates the branch; no further state change.
- **P2** (propagation): each of REFL/TRANS/MON adds a relational atom or formula placement to  $\mathcal{R}$  or  $\Gamma$ ;  $\triangleright R_2$  adds a formula to  $\Delta$ . Each fires only when its trigger is unprocessed.
- **P3** (branching): creates child branches; each child inherits the current state and continues with at least one new formula placement per child.
- **P4/P5** (generation): either creates a new label (strictly increasing  $\mathcal{L}$ ) or triggers a redirect.

*Redirect cascades.* When redirect fires (Definition 6.14, step 3), the provisional label  $y$  is removed and its obligations are transferred to  $z$ . Steps 3b–3e may or may not enlarge the material components (if every transferred atom, formula, and annotation is already present at  $z$ , the material state does not grow). However, the generating trigger that created  $y$  is added to  $\mathcal{P}$ , and each subsequent P2 application triggered by step 3e processes its own fresh trigger, so  $\mathcal{P}$  grows at every step of the cascade. Since  $|\mathcal{P}| \leq T$ , the cascade terminates.

The total number of steps *per branch* is therefore at most  $T = O(N^3 + N^2n) = 2^{O(n)}$ , since  $N = (d+1) \cdot 3^n$  is already singly exponential in  $n$ . (Each branch maintains its own processed-trigger set  $\mathcal{P}$ ; branching rules create child branches that each inherit the parent's state and continue independently. The bound  $T$  is uniform across all branches because every branch operates within the same trigger space.)  $\square$

**Proposition 6.21** (Countermodel construction). *If the proof-search procedure yields a saturated open branch  $\mathcal{B}$ , then  $\mathcal{M}_{\mathcal{B}}$  is an  $S$ -serial bare model falsifying  $\varphi_0$ .*

*Proof.* We verify that the extracted structure  $\mathcal{M}_{\mathcal{B}}$  (Definition 6.7) is an  $S$ -serial bare model.

*Preorder.*  $\leq$  is reflexive by (Sat-refl) and transitive by (Sat-trans).

*Monotone valuation.* If  $x \in V(p)$  and  $x \leq y$ , then  $x:p \in \Gamma$ , and by (Sat-mon)  $y:p \in \Gamma$ , so  $y \in V(p)$ .

*Forward confluence.* If  $x \leq y$  and  $xRz$ , then by (Sat-fc)—applied at all labels, since no label is blocked—there exists  $u$  with  $yRu$  and  $z \leq u$ .

*S-seriality.* For every label  $x$ , (Sat-ser)—applied at all labels—gives  $y$  with  $xRy$  and  $x \leq y$ , i.e.,  $xSy$ .

Since all saturation conditions (P2–P5) hold at *every* label, the truth lemma (Lemma 6.8) applies directly at all labels. From  $x_0:\varphi_0 \in \Delta$  we obtain  $\mathcal{M}_{\mathcal{B}}, x_0 \not\models \varphi_0$ .  $\square$

**Theorem 6.22** (Completeness). *If  $\varphi$  is valid in all  $S$ -serial bare models, then  $\varphi$  is provable in **LS-DWS**.*

*Proof.* Suppose  $\varphi$  is not provable. We construct a falsifying model by systematic proof search.

*Global search structure.* Initialise the root sequent with a single label  $x_0$  and  $x_0:\varphi \in \Delta$ . The proof-search procedure of Definition 6.10 builds a *proof tree* by applying rules in priority order. Branching occurs only at P3 rules ( $\wedge R$ ,  $\vee L$ ,  $\rightarrow L$ ,  $\triangleright L$ ), each of which produces finitely many immediate successors (at most two). All other rules (P1, P2, P4, P5 and redirect) extend the current branch without branching. Therefore every node in the proof tree has finite degree.

*From non-provability to a saturated open branch.* By Lemma 6.20, every branch of the proof tree terminates in finitely many steps. Since every node has finite degree, the proof tree is finitely branching. Moreover, Lemma 6.20 provides a *uniform* bound: every branch has at most  $T$  steps. If the proof tree were infinite, König’s lemma would yield an infinite branch, contradicting this bound. Therefore the proof tree is finite.

If every leaf of this finite tree were closed (i.e., axiom), the tree would constitute a derivation of  $\varphi$  in **LS-DWS**, contradicting our assumption. Hence at least one leaf is an open branch  $\mathcal{B}$ . This branch is *saturated*: the algorithm halts at a leaf only when no unprocessed trigger remains (Definition 6.10), so every applicable rule has been applied. Proposition 6.21 then extracts from  $\mathcal{B}$  an  $S$ -serial bare model falsifying  $\varphi$ .  $\square$

## 7 Decidability

**Theorem 7.1.** *Validity of  $\mathcal{L}_{\text{DWS}}$ -formulas over  $S$ -serial bare frames is decidable.*

*Proof.* By Theorem 6.22 and Lemma 6.20.  $\square$

**Corollary 7.2** (Bounded model property). *If  $\varphi_0 \in \mathcal{L}_{\text{DWS}}$  is falsifiable, it is falsifiable in an  $S$ -serial bare model of size at most  $(d+1) \cdot 3^n$ , where  $n = |\text{Sub}(\varphi_0)|$  and  $d$  is the modal depth of  $\varphi_0$ . In particular the bound is singly exponential.*

*Proof.* The countermodel extracted in Proposition 6.21 has at most  $N$  worlds, and  $N \leq (d+1) \cdot 3^n$  by Lemma 6.18.  $\square$

**Corollary 7.3** (Finite model property). *DWS has the finite model property: every falsifiable formula is falsifiable in a finite  $S$ -serial bare model.*

*Proof.* The countermodel of Proposition 6.21 is finite (its universe is the finite set of labels).  $\square$

*Remark 7.4.* The  $(d+1) \cdot 3^n$  bound is not claimed to be tight. We conjecture PSPACE-completeness. A PSPACE lower bound follows from the PSPACE-hardness of intuitionistic propositional logic [9], which embeds into DWS via the  $\triangleright$ -free fragment; a PSPACE upper bound would require a polynomial-space algorithm that avoids materialising the full label graph. The  $3^n$  factor in the current bound is an artefact of the global type registry, which memoises types across the entire search space. A depth-first strategy that reuses space across branches—materialising only a single root-to-leaf path of polynomial length at a time, in the style of Ladner’s algorithm for classical modal logics or Hudelmaier’s contraction-free calculus for intuitionistic logic—is the standard path to a PSPACE upper bound.

## 8 Comparison with Intuitionistic Conditional Logic

Weiss [10] introduced ICK; Olkhovikov [6] introduced IntCK with a might-conditional  $\Diamond \rightarrow$ ; Dalmonte and Girlando [1] developed nested and sequent calculi for intuitionistic conditional logics, including IntCK. Their approach uses selection-function terms directly in labels, whereas

our annotated-challenger mechanism avoids explicit selection functions by tracking falsification witnesses.

**Definition 8.1** (IntCK frame). An *IntCK frame* is  $\langle W, \leq, f \rangle$  where  $\leq$  is a preorder and  $f: \mathcal{P}(W) \times W \rightarrow \mathcal{P}(W)$  is a selection function satisfying *monotonicity*: if  $w \leq w'$ , then for every  $u \in f(X, w)$  there exists  $u' \in f(X, w')$  with  $u \leq u'$ . The might-conditional has clause:  $w \models \varphi \Diamond \rightarrow \psi$  iff  $\forall v \geq w (v \models \varphi \Rightarrow \exists u \in f(\llbracket \varphi \rrbracket, v) (u \models \psi))$ .

**Definition 8.2** (Constant-selection-function embedding). Given an  $S$ -serial bare frame  $\mathcal{F} = \langle W, \leq, R \rangle$ , define  $f_{\mathcal{F}}(X, v) = \{u : vSu\}$  for all  $X, v$ . Note that  $f_{\mathcal{F}}$  ignores its first argument, so  $\triangleright$  behaves as an *intuitionistic strict existential implication* over  $S$ : the witness set  $\{u : vSu\}$  depends only on the evaluation world, not on the antecedent.<sup>6</sup> See Remark 8.4 for the consequences regarding monotonicity.

**Theorem 8.3** (Pointwise correspondence). *Under the constant-selection-function embedding (see Remark 8.4 for the monotonicity caveat), for every model and world:  $\mathcal{M}, w \models \varphi \triangleright \psi$  iff  $\mathcal{M}^*, w \models \varphi \Diamond \rightarrow \psi$ , where  $\mathcal{M}^*$  uses selection function  $f_{\mathcal{F}}$ .*

*Proof.* Both clauses reduce to  $\forall v \geq w (v \models \varphi \Rightarrow \exists u (vSu \wedge u \models \psi))$ .  $\square$

*Remark 8.4.* The constant selection function  $f_{\mathcal{F}}$  may fail the IntCK monotonicity condition (Definition 8.1), since this condition amounts to  $S$ -forward confluence, which fails in general (Proposition 3.1). Theorem 8.3 is a *pointwise* semantic equivalence of satisfaction clauses: for each model and world, the two connectives have the same truth value. Persistence of  $\triangleright$  in DWS is ensured by the implication wrapper (Theorem 2.3), not by selection-function monotonicity. Frame-level transfer of validities between the two systems requires the additional assumption of  $S$ -forward confluence: if a frame satisfies  $S$ -forward confluence, then  $f_{\mathcal{F}}$  is monotone and the frame embeds into the class of IntCK frames, so every IntCK validity holds under the embedding. The pointwise correspondence (Theorem 8.3) does not by itself yield transfer of validities to IntCK, because without  $S$ -forward confluence the induced constant selection function may fail the IntCK monotonicity condition, so the induced structures need not belong to the IntCK frame class. The four-world frame of Proposition 3.1 witnesses this: the biconditional  $\varphi \triangleright \psi \leftrightarrow \varphi \Diamond \rightarrow \psi$  holds at every world in every model on this frame, yet the frame itself falls outside the IntCK class, preventing global transfer.

**Proposition 8.5** (Antecedent independence). *The following is a key structural advantage of DWS over general IntCK: the witness set  $\{u : vSu\}$  is independent of the antecedent, so conduction distributes over disjunction. Over  $S$ -serial bare frames:  $\models (\varphi_1 \triangleright \psi) \wedge (\varphi_2 \triangleright \psi) \leftrightarrow ((\varphi_1 \vee \varphi_2) \triangleright \psi)$ .*

*Proof. Right-to-left.* If  $w \models (\varphi_1 \vee \varphi_2) \triangleright \psi$  and  $v \geq w$  with  $v \models \varphi_i$ , then  $v \models \varphi_1 \vee \varphi_2$ , so  $\exists u (vSu \wedge u \models \psi)$ .

*Left-to-right.* If  $w \models (\varphi_1 \triangleright \psi) \wedge (\varphi_2 \triangleright \psi)$  and  $v \geq w$  with  $v \models \varphi_1 \vee \varphi_2$ : by cases on which disjunct holds, the corresponding conjunct provides  $\exists u (vSu \wedge u \models \psi)$ . The witness set  $\{u : vSu\}$  is independent of the antecedent.  $\square$

**Example 8.6.** Antecedent independence fails in IntCK with non-constant selection functions: if  $f(\llbracket \varphi_1 \rrbracket, w) \neq f(\llbracket \varphi_2 \rrbracket, w)$ , the witness sets for the two antecedents differ, and  $f(\llbracket \varphi_1 \vee \varphi_2 \rrbracket, w)$  need not contain a  $\psi$ -world.

The formula  $\top \triangleright \psi$  defines the persistent analogue of  $\langle S \rangle \psi$ :  $w \models \top \triangleright \psi$  iff  $\forall v \geq w \exists u (vSu \wedge u \models \psi)$ .

<sup>6</sup>This antecedent independence is the algebraic reason why  $\triangleright$  distributes over disjunction in its first argument (Proposition 8.5), in contrast to variably strict conditionals in the Lewis-Stalnaker tradition, where the selection function depends on the antecedent proposition.

## 9 Conclusion and Open Problems

We have introduced conduction  $\triangleright$ , proved irreducibility over  $\mathbf{IK}$ , provided the Hilbert system **H-DWS** with soundness, proved completeness, decidability, and the finite model property for **LS-DWS** with a singly exponential bound (at most  $(d+1) \cdot 3^n$  labels), and established a formal correspondence with IntCK.

*Hilbert completeness.* Is **H-DWS** complete for  $S$ -serial bare frames? The standard canonical-model strategy for intuitionistic modal logics (as in Simpson [8]) constructs a preorder  $\leq^c$  on prime theories and an accessibility relation  $R^c$  satisfying forward confluence. For DWS, completeness requires the additional property  $S^c$ -seriality: every prime theory  $\Gamma$  must have an  $S^c$ -successor, i.e., some  $\Delta$  with  $\Gamma R^c \Delta$  and  $\Gamma \leq^c \Delta$ . The difficulty is that the canonical  $R^c$  (built from  $\Box/\Diamond$ -witnesses) and the canonical  $\leq^c$  (set inclusion of prime theories) are constructed independently, and their intersection need not be serial: a world  $\Gamma$  may have  $R^c$ -successors none of which extend  $\Gamma$ . The conduction axioms provide  $S$ -witnesses *conditionally* (given  $v \models \varphi$ ), but the unconditional seriality of  $S^c$  at  $\Gamma$  does not obviously follow. An approach via neighbourhood semantics, where the selection function is built directly from the conduction axioms, may bypass this obstacle: in a neighbourhood model the  $S$ -successor sets are given explicitly rather than computed as the intersection of independently constructed  $R$  and  $\leq$ , so  $S$ -seriality can be enforced at construction time by stipulating a non-empty neighbourhood for every world. The challenge is to verify that the resulting neighbourhood structure validates the full axiomatisation and not merely the conduction fragment.<sup>7</sup>

A closely related question: is **H-DWS** proof-theoretically conservative over  $\mathbf{IK}_S$  for  $\mathcal{L}_{\mathbf{IK}}$ -formulas? Since Hilbert completeness is open, this cannot currently be resolved; we record it as an explicit open problem.

*Cut elimination.* The annotated-challenger mechanism introduces non-locality; standard cut-elimination strategies may require adaptation.

*Complexity.* We conjecture PSPACE-completeness.

*DWS without seriality.* Without  $S$ -seriality, axiom C1 fails. We conjecture that the resulting logic is axiomatised by  $\Box(\varphi \rightarrow \psi) \wedge (\top \triangleright \top) \rightarrow (\varphi \triangleright \psi)$ , conditioning on local  $S$ -seriality via  $\top \triangleright \top$ .

*Further.* Does DWS enjoy Craig interpolation? What is the algebraic counterpart? What is the minimal frame condition for  $S$ -forward confluence without  $R \subseteq \leq$ ?

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<sup>7</sup>A filtration argument over a finite set of formulas might also circumvent the infinite prime-theory intersection issue, since filtration can enforce  $S$ -seriality by construction on a finite quotient. However, without a clear subformula property for the axiomatic system, it is not obvious how to define the appropriate filtration.

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